



ARTICLE



<https://doi.org/10.1057/s41599-025-05817-5>

OPEN

AI-driven language learning in higher education: an empirical study on self-reflection, creativity, anxiety, and emotional resilience in EFL learners

Min He^{1,2✉}, Babar Nawaz Abbasi³ & Jinhua He⁴

In modern times, artificial intelligence (AI) technologies are reshaping traditional methods of language learning in higher education. However, their impact on language learners' self-reflection, creativity, anxiety reduction, and emotional resilience has not been empirically explored in depth. Therefore, this study aims to fill that gap by examining English as a Foreign Language (EFL) learners using Structural Equation Modeling (SEM), Quantile Regression (QR), and Phenomenological Analysis (PA), using a survey of 205 EFL undergraduate learners through semi-structured and structured questionnaires from various Chinese universities. The results revealed that corrective AI-powered feedback, such as grammar and vocabulary corrections, along with motivational feedback, including encouragement and progress tracking, significantly improved EFL learners' self-reflection. Furthermore, creativity and AI-powered feedback encouraged EFL learners to become more creative in their use of English, more confident in expressing original ideas, and increased enjoyment in writing and speaking. Moreover, familiarity with AI-powered feedback and the manner in which EFL learners receive this feedback have partially reduced their performance anxiety. In addition, AI-powered feedback is enhancing emotional resilience by making them more confident in overcoming setbacks. However, the study offers policy suggestions for both government and AI vendors/companies.

¹ School of International Studies, Shaanxi Normal University, Xi'an, China. ² Xi'an Artificial Intelligence and Second Language Acquisition International Science and Technology Cooperation Base, Xi'an, China. ³ Institute of China Innovation and Entrepreneurship Education, Hangzhou Normal University, Hangzhou, China. ⁴ School of Mathematics, Southwest Jiaotong University, Chengdu, China. ✉email: hemin@snnu.edu.cn

Introduction

Artificial intelligence (AI) is rapidly transforming language learning in higher education, offering personalized feedback, adaptive content, and real-time interaction that enhance student engagement and fluency (Sajja et al. 2024). Recent studies highlight how AI integration in EFL classrooms fosters more dynamic and student-centered learning environments. For instance, Wang et al. (2025) emphasize the importance of classroom climate, AI literacy, and student resilience in promoting engagement within AI-assisted Chinese EFL settings. Similarly, Gao et al. (2024) explore how university EFL teachers in China are adapting their pedagogical beliefs to integrate tools like ChatGPT and other large language models into instruction. However, as learners increasingly rely on digital platforms to support their academic growth, these tools are not only reshaping instructional methods but also influencing learners' emotional and psychological experiences. The growing usage of AI-powered language tools as part of educational technology has prompted several scholars throughout the world to investigate the benefits and drawbacks of AI-powered tools in various language courses, particularly in higher education. However, the impact of AI-powered feedback on key learner variables, including self-reflection, creativity, performance anxiety, and emotional resilience among EFL learners, has not gotten the attention it merits despite its influence on the learning of EFL in various countries. In essence, there is a scarcity of studies specifically conducted to uncover the impact of AI-powered feedback on the behavioral elements of EFL learners in higher education. Likewise, investigating whether there is a balance between AI integration and the cognitive development of EFL learners, i.e., whether AI-powered feedback is upgrading or downgrading the creativity of EFL learners, remains a subject of debate. Similarly, examining whether AI-powered tools reduce the performance anxiety of EFL learners has not been explored to the fullest; however, there are relevant studies, such as Jafari et al. (2015), Chen et al. (2022), Karimi et al. (2022), Huang and Tan (2023), and Al Rashidi and Aberash (2024). Furthermore, the integration of AI with EFL learning is still a developing field, and debates are in their early stages in higher education.

A large size of literature has portrayed the rapid advancement of AI-powered technologies to benefit language learning, specifically English as a foreign language, as it has introduced transformative possibilities across various sectors. In recent years, linguists and educators have increasingly explored the integration of AI technologies in language learning, specifically EFL acquisition (Lei et al. 2022). AI tools, such as personalized learning systems and intelligent feedback mechanisms, have demonstrated significant potential to revolutionize traditional pedagogical approaches (Fathi and Rahimi, 2022). Furthermore, the integration of AI-powered feedback into learning environments offers personalized, interactive, and communicative benefits that go beyond traditional classroom settings. AI-driven systems enable learners to receive tailored feedback that not only corrects errors but also motivates them, fostering deeper engagement with the learning material. This shift toward AI-enhanced learning allows educators to better understand individual learners' strengths and weaknesses, adapting teaching strategies accordingly.

However, while AI-powered feedback offers significant advantages, some challenges remain. For instance, overreliance on AI-driven tools may stifle critical thinking and creativity, as both learners and educators may become overly dependent on automated systems for language acquisition (Wang et al., 2024). Furthermore, the integration of AI technologies in EFL education poses potential barriers, such as inadequate technological proficiency among teachers and students, which can hinder effective use of these tools (Zou et al. 2020). Moreover, the lack of well-

designed interactive exercises in some AI systems limits their ability to fully engage learners and reduce anxiety. Despite these challenges, AI-powered technologies hold significant promise for the future of language education, offering tools that can enhance motivation, improve learner outcomes (AlTwijri and Alghizzi, 2024). Likewise, it is evident from the literature that the techniques and approaches employed to evaluate such phenomena are primarily focused not on causal relationships but on response analysis of individual variables, which may not provide the required insights for policy construction or the development of EFL learning. Thus, the analyses are often based on descriptive and simple inferential methods, such as mean and standard deviation, frequency distribution, ANOVA, ANCOVA, MANOVA, T-tests, and similar approaches. Advanced estimation techniques that could provide more reliable information for policy formulation and development are not being applied. Hence, empirical and methodological gaps exist, which serve as the inspiration to carry out this study to address these gaps. Therefore, in an effort to fill these gaps, this study ponders to investigate AI-driven language learning in higher education with respect to self-reflection, creativity, anxiety, and emotional resilience in EFL learners with the following objectives:

- i. To investigate whether the type of AI-powered feedback (e.g., corrective vs. motivational) causally affects self-reflection processes in EFL learners;
- ii. To examine the causal relationship between the use of AI-powered feedback and the development of creativity in EFL learners;
- iii. To explore the extent to which AI-powered feedback reduces performance anxiety in EFL learners, and how mediating variables (e.g., learner's familiarity with AI, feedback delivery style) influence this relationship; and
- iv. To analyze whether the integration of AI-powered feedback in language learning affects the long-term improvements in EFL learners' emotional resilience.

The innovations of this study as its contributions to the existing literature are as follows. First, it uncovers the impact of AI-powered feedback on the behavioral elements of the learners of EFL, particularly the influence of AI-powered feedback on enhancing the self-reflection of the EFL learners. Second, investigating whether there is a balance between the AI integration and cognitive development of the EFL learners, i.e., whether the AI-powered feedback is upgrading or downgrading the creativity of the learners of EFL. Third, examining whether the AI-powered tools are reducing the performance anxiety of the EFL learners in collaboration with AI-powered feedback. Fourth, this study advances methodological rigor by integrating multiple analytical techniques. While SEM and PA are employed as primary analytical methods, PCA is used to construct an index from latent variables, transforming them into an observable variable comparable to a time-series variable. To further ensure the robustness of the findings, Quantile regression (QR) is applied to validate the estimates obtained from SEM, thereby enhancing the study's reliability and credibility. Nonetheless, the study's conceptual depiction that visually represents the relationships between AI-powered feedback types (corrective vs. motivational) and key learner outcomes such as self-reflection, creativity, emotional resilience, and performance anxiety is shown in Fig. 1.

Literature Review

Theoretical review

The role of self-reflection in learning. The capacity to observe and assess one's own mental, emotional and behavioral processes is

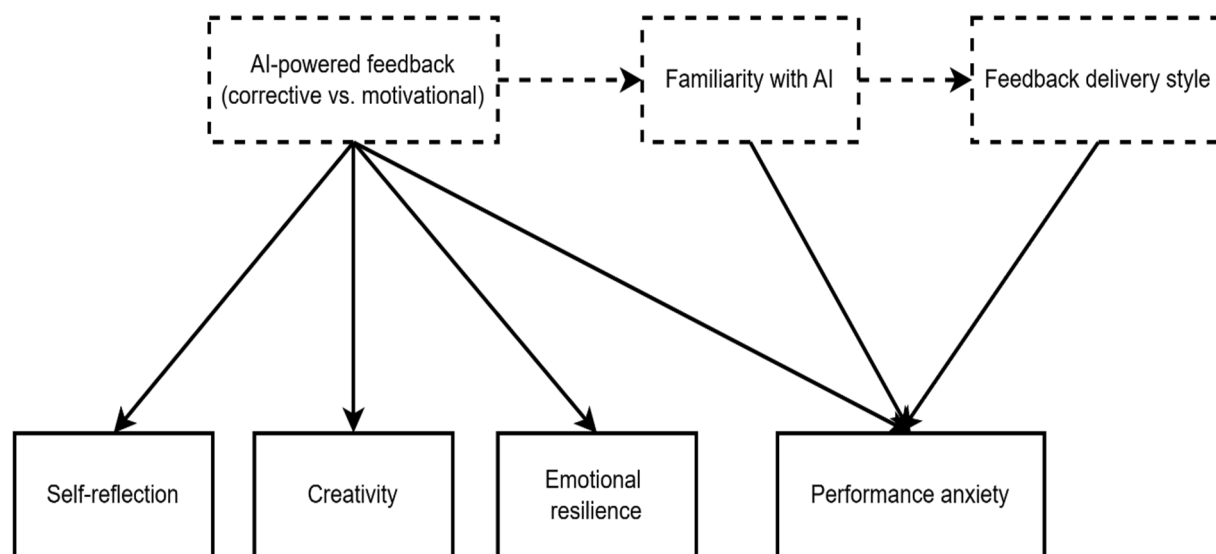


Fig. 1 Conceptual framework of the study.

known as self-reflection (Dishon et al. 2017). Self-development also heavily relies on self-reflection. Assessing your strengths and weaknesses, or what you did well and what you did poorly, can help you pinpoint areas that need improvement so that you can address them (Nowak et al. 2022). Reflective thinking is the most important skill in fostering learning in complex problem-solving situations because it allows students to stand back and think about how to solve the problem and how a set of problem-solving techniques is used to achieve their goals (Akpur, 2020; Orakci, 2021). Students who participate in consistent and structured reflection demonstrate increased self-awareness and understanding of their optimal learning methods. They possess enhanced capabilities to manage the learning process and maintain motivation for learning (Wordpress, 2023). Recent studies highlight that self-reflection plays a pivotal role not only in academic learning but also in emotional intelligence development. It aids in the emotional adjustment of learners to various academic pressures, thereby enhancing their overall educational resilience (Marvell and Simm, 2021). According to Erdogan (2019), educators have started to integrate reflective practices into the curriculum, which has significantly boosted students' critical thinking skills, allowing them to better integrate theoretical knowledge with practical applications. Moreover, technology-enhanced reflective practices, such as digital portfolios and online reflective journals, have gained traction. These tools provide students with platforms to systematically reflect on their learning experiences and track their progress over time. This digital approach to reflection has been particularly effective in remote and blended learning environments, where direct teacher feedback and peer interaction may be less frequent (Mumford and Dikilitaş, 2020).

Creativity in language use. Creativity refers to the inclination to produce or identify ideas, alternatives, or possibilities that can be beneficial in addressing problems, facilitating communication, and providing entertainment for oneself and others. Relatively recent papers (for example, Jones, 2016; Jones and Richards, 2015; Maley and Kiss, 2017) on the topic of creativity demonstrate that creativity is becoming an increasingly important topic in the field of applied linguistics research. According to Mohebbi and Coombe (2021), this trend is commensurate with educational frameworks of the 21st century that place an emphasis on innovation. This research domain encompasses a wide range of

knowledge, which has been shaped by the theoretical and empirical insights of linguists, psychologists, and educational theorists. Some of these individuals include Rhodes (1961), Csikszentmihalyi (1990), Amabile (1996), and Johnson (2010), amongst others. The goal of this research domain is not to be exhaustive. Research on creativity has primarily focused on the language elements rather than the practical dimensions in the field of applied linguistics. This is because the linguistic aspects are more easily understood. In the context of ordinary language use, research on linguistic creativity reveals that it is a trait that is shared by all people, rather than being a characteristic that is exclusive to exceptional individuals. Carter (2015) asserts, "linguistic creativity is not simply a property of exceptional people but an exceptional property of all people." Further exploring this concept, studies such as those by Sawyer and Henriksen (2024) and Beghetto (2007) have investigated how individuals use creativity in everyday language to solve problems and enhance communication. These works emphasize the role of creativity not only in artistic contexts but also in daily interactions and teaching methods. Moreover, Scott et al. (2004) highlights the importance of the environment in fostering creative use of language, suggesting that supportive contexts can significantly enhance creative outputs in language learning. In addition, the influence of digital platforms on linguistic creativity has been explored by Crystal (2011), who notes that technology and internet communication are reshaping traditional notions of creativity in language.

Anxiety in language learning. Anxiety is a state of apprehension, apprehension, and unease that is influenced by both internal and external factors that are pertinent to foreign language learners. During the process of acquiring a second language, students may experience anxiety, which can lead to distress in their performance. According to Brown (2000), three emotions in foreign language learning performances: communication apprehension, which is defined as learners' difficulty in articulating ideas and thoughts; test anxiety; and apprehension of negative evaluation, which arises from the necessity of maintaining a favorable social image. Anxiety adversely affects language acquisition and has emerged as one of the most well-investigated factors in the psychological dimensions of education (Horwitz, 2001). Therefore, anxiety causes English learners to experience a complete incapacity to speak English, owing to anxiety, usually known as a "mental block" in language acquisition. Furthermore, MacIntyre

and Gardner (1994) discuss how anxiety can interfere with the processing of new information and negatively impact the retention and production of the new language. Their studies reveal that higher levels of anxiety correlate with lower performance in language tasks. Moreover, recent meta-analyses by Gkonou et al. (2017) confirm that anxiety not only reduces spoken output but also affects learners' willingness to communicate, thereby stifling language development over time. In fact, AI-powered feedback mechanisms are particularly beneficial for EFL learners, as they promote self-reflection, encourage creativity, and equip learners with tools to manage performance anxiety (Bakare and Jatto, 2023). These tools allow learners to reflect on their progress, identify areas for improvement, and experiment with new language structures in a supportive environment (Xu et al. 2023).

Emotional resilience in language learning. Emotional resilience, defined as an individual's ability to adapt to challenges, stress, and setbacks, plays a significant role in language learning (Dewaele et al. 2019). Foreign language acquisition often involves uncertainty, errors, and communication anxiety, making emotional resilience a crucial factor in learner success. Studies suggest that learners with higher emotional resilience are better able to persist in language learning, develop positive attitudes, and manage performance-related anxiety (MacIntyre and Gregersen, 2012). Furthermore, the process of acquiring a new language often involves linguistic uncertainty, which can lead to frustration and reduced motivation. Resilient learners are more likely to embrace challenges, persist through difficulties, and maintain positive attitudes toward learning (Dewaele and MacIntyre, 2016). Oxford (2016) highlights that emotional resilience is closely tied to self-regulated learning strategies, such as self-efficacy, intrinsic motivation, and goal-setting, all of which enhance long-term language proficiency. Xue (2022) further highlights that EFL teachers' emotional resilience and self-efficacy are also instrumental in fostering learners' academic success, suggesting that resilience is not only a learner trait but also a vital characteristic of effective educators. Similarly, Wang et al. (2024) provided insights into how emotional resilience mitigates burnout and promotes engagement among Chinese high school EFL learners, reaffirming resilience as a key psychological buffer in demanding academic settings. Recent findings by Qi and Derakhshan (2024) also reveal that technology-based collaborative learning enhances learners' social regulation skills, positively influencing their emotional states and academic performance. This suggests the socio-emotional benefits of well-structured digital learning environments. Recent research indicates that AI-assisted learning environments can support emotional resilience by providing personalized feedback, reducing anxiety, and fostering self-reflection (Vera, 2023). AI-powered tools like chatbots and adaptive learning systems encourage autonomous learning, allowing students to progress at their own pace and reduce fear of failure (Kim et al. 2023). Students' English learning resilience mediated the relationship between English learning burnout and English academic achievement (Liu et al. 2024). AI-driven feedback enhances learner confidence, which is critical in reducing communication anxiety and promoting engagement in language acquisition (Yang, 2024).

AI in English learning. Learners of English regularly make use of AI tools, such as Ask AI, ChatGPT, OpenAI, and Perplexity, in addition to other educational technologies, in order to acquire new information, comprehend instructional presentations, and remember lessons (Warschauer et al. 2023). The use of AI in education, which includes intelligent tutoring systems, language learning applications, and adaptive learning platforms, has attracted a significant amount of attention from researchers,

educators, and policymakers all over the world (Ilkka, 2018; Kim, 2019; Chen et al. 2022; Huang and Tan, 2023; Xia et al. 2022; Zhang and Zou, 2020). Researchers have found that AI is being used extensively in the field of language acquisition to improve the linguistic competencies and sub-competencies of learners. ChatGPT is a tool that is supported by AI that may be used in language learning contexts to assist learners in increasing their language competencies and sub-competencies (Fang et al. 2023; Fitria, 2023; Kim, 2023; Schmidt-Fajlik, 2023; Su et al. 2023; Warschauer and Xu, 2024; Yan, 2023). AI has become a prominent force in EFL education, transforming the way language instruction is delivered and how learners engage with content (Fan and Zhang, 2024). Liu and Fan (2024) argued that AI literacy significantly mediates the relationship between teacher support, technical support, and learners' willingness to communicate. In other words, when learners possess the skills to effectively understand and use AI tools, they are more likely to benefit from the support structures around them and demonstrate greater communicative confidence.

Researchers, such as Xu et al. (2023), Hwang et al. (2023), and Su et al. (2014), have found that AI is being used extensively in the field of language acquisition to improve the linguistic competencies and sub-competencies of learners. AI-powered tools, such as ChatGPT, play a transformative role in the language learning process by providing instant, personalized feedback and real-time interaction, which are crucial for mastering a language. For instance, these tools allow learners to practice speaking, writing, and listening skills by engaging in simulated conversations and exercises, making language learning more dynamic and less dependent on traditional classroom methods. Furthermore, AI systems can analyze a learner's specific errors, such as grammar or pronunciation mistakes, and offer tailored corrections that enable more targeted practice.

Empirical review. Existing studies on the use of AI-powered feedback in EFL education reveal its significant potential to enhance learner engagement and proficiency (Amin, 2023; Kukulska-Hulme and Viberg, 2018; Shen et al. 2023). These studies have shown that AI-assisted feedback tools can improve accuracy and fluency in language use by providing real-time corrections and suggestions tailored to individual learning paces and styles. Furthermore, these tools promote creativity and self-reflection among learners by offering personalized insights and challenges that adapt to their specific developmental stages. However, a notable gap remains in understanding the impacts of such technology on emotional resilience. While positive influences are broadly recognized, the depth of emotional responses and their effects on sustained language learning engagement are underexplored. This gap signifies the need for more detailed studies that investigate emotional outcomes and consider the diverse cultural backgrounds of EFL learners to optimize AI feedback systems for global educational contexts. The empirical review is divided into two sections. The first section examines empirical studies related to the relationship between EFL and learners' behavioral factors. The second section focuses on empirical studies concerning the relationship between AI-powered feedback and the behavioral factors of EFL learners.

Nexus between EFL and learners' behavioral factors. This section reviews empirical studies investigating how various behavioral factors, such as motivation, engagement, and self-regulation, influence and are influenced by EFL learning. Research in this area often examines the dynamics between learners' psychological traits and their ability to acquire language proficiency effectively. Factors like learner autonomy, cultural attitudes, and classroom environment play a pivotal role in shaping the outcomes of EFL

instruction. For example, Sarani et al. (2014) examined the impact of self-evaluation on the accuracy and fluency of English speaking among intermediate and upper-level learners. The study utilized a sample of 30 pre-intermediate and 30 upper-intermediate students, employing an ANCOVA test for analysis. The findings indicated that self-evaluation had a positive effect on participants' speaking accuracy and fluency. Jafari et al. (2015) explored the impact of self-evaluation on the language proficiency and ambiguity tolerance of a sample of 20 Iranian intermediate EFL learners. The results indicated that self-evaluation enhances learners' language proficiency and ambiguity tolerance. Karimi et al. (2022) conducted a survey involving 63 EFL learners to assess the effectiveness of reflective thinking in enhancing meta-cognitive awareness and subsequently improving reading comprehension, utilizing the MANOVA test. The findings indicated that reflective thinking influenced the assessments of EFL students. Al Rashidi and Aberash (2024) investigated the impact of reflective thinking and self-evaluation in language acquisition on EFL students' development of mindfulness, resilience, and academic well-being, utilizing a sample of 96 intermediate students and adopting One-way ANOVA and Tukey tests. The results indicate that self-assessment and reflective cognition are crucial elements in the acquisition of EFL. Yaghoobi and Razmjoo (2016) explored the potentiality of computer-assisted instruction towards ameliorating Iranian EFL learners' reading level. The study used a sample of 50 EFL learners, split into experimental and control groups. Results from an independent t-test showed that learners using computer-assisted instruction significantly improved their reading skills compared to those receiving traditional instruction. The findings suggest that CAI is more effective in enhancing reading comprehension in Iranian EFL learners.

AI-powered feedback on the behavioral factors of EFL learners. This section explores empirical studies focusing on the integration of AI-powered feedback tools in EFL education and their effects on learners' behavioral factors. This includes changes in motivation, self-confidence, engagement, and adaptation to personalized learning environments. The research highlights the transformative role of AI in providing immediate and tailored feedback, fostering deeper learner involvement and improving overall proficiency through behavioral engagement. For instance, Dai and Liu (2024) examined leveraging AI and EFL class: challenges and opportunities in the spotlight using a phenomenological approach. The study gathered data from 45 Chinese EFL students through open-ended questionnaires and interviews, analyzing the results with MAXQDA software. Findings revealed that AI in EFL classes offers benefits like individualized learning and immediate feedback but also presents challenges for students. The study provides insights for educators to mitigate the difficulties of AI implementation in language classrooms. Syifaiddin and Yuliansyah (2023) analyzed the impact of AI on the motivation and anxiety of EFL students in learning English, utilizing a sample of 51 students and employing a T-test for their analysis. The findings indicated that the incorporation of AI in English language learning markedly enhanced student motivation. Wei (2023) investigated the effects of AI on the achievement, motivation, and self-regulation of EFL learners in China, utilizing a sample of 60 students and employing mixed-design ANOVA and quantile-quantile (Q-Q) plots for analysis. The findings indicate that AI-mediated instruction enhances learning achievement, motivation, and self-regulated learning among EFL learners. Marzuki et al. (2023) evaluated the effects of AI writing tools on student writing as perceived by EFL teachers, employing a qualitative methodology with a sample of four EFL teachers. The findings suggest that the incorporation of AI writing tools may enhance the quality of writing among EFL students. Furthermore,

Yuan and Liu (2024) explored the effect of AI tools on EFL learners' engagement, enjoyment, and motivation using a quasi-experimental approach. The study involved 383 Chinese EFL learners, divided into experimental (Duolingo users) and control groups, with pre- and post-intervention assessments across 12 sessions. One-way ANCOVA analysis revealed significant improvements in engagement, motivation, and foreign language enjoyment in the experimental group. Findings highlight the value of AI tools like Duolingo in enhancing engagement and motivation in EFL classrooms. Guo et al. (2024) examined the effect of an AI-supported approach to peer feedback on the quality of feedback and writing skills of EFL students, utilizing a sample of 124 participants and applying Quade's test. The findings indicated that AI integration improved the quality of student feedback and their writing abilities. Moreover, Selim (2024) explored the impact of AI on the academic writing skills of university-level EFL students, utilizing a sample of 50 students and employing a mixed-methods approach. The findings indicated that EFL students recognize a positive effect on writing quality, yet they express uncertainty regarding improvements in confidence. Woo et al. (2024) investigated the impact of AI-generated text on the writing of 23 Hong Kong secondary school EFL students. Analysis revealed that both human and AI-generated text improved writing scores, with AI benefiting both high- and low-scoring students. The study suggests tailored strategies for EFL students to use AI tools effectively. Alzahrani and Alotaibi (2024) conducted the influence of AI, specifically ChatGPT, on the writing skills of high school students with intermediate-level EFL proficiency, utilizing paired-sample t-tests for analysis. The findings indicated enhancements in various writing competencies, including task achievement, coherence and cohesion, and lexical resource. Wang (2024) investigated the impact of corrective feedback on writing anxiety among language learners using an AI-driven application named Poe, analyzing a sample of 25 learners through ANOVA testing. The findings indicated that participants receiving AI-generated feedback exhibited a greater decrease in writing anxiety compared to their counterparts. Zhang et al. (2024) examined the impact of AI on EFL speaking through a quasi-experimental study involving 131 students. The findings underscored the beneficial impact of AI-speaking assistants on EFL students' enjoyment of foreign language learning, reduction of foreign language anxiety, and increased willingness to communicate in English. Wang and Xue (2024) explored using AI-driven chatbots to foster Chinese EFL students' academic engagement in a quasi-experimental study. The results demonstrated that AI chatbots significantly enhanced student engagement by providing interactive feedback and creating a supportive learning environment.

Literature gaps. A review of existing studies reveals that the impact of AI-powered feedback on key behavioral elements of EFL learners such as self-reflection, creativity, and performance anxiety, has not been sufficiently explored (Jafari et al. 2015; Chen et al. 2022; Karimi et al. 2022; Huang and Tan, 2023; Al Rashidi and Aberash, 2024). While these factors are recognized for their influence on language learning, the literature lacks focused investigations into how AI-powered feedback affects these specific behavioral outcomes. Furthermore, there is limited research addressing the long-term effects of AI feedback on EFL learners' emotional resilience and its broader implications for language education. Existing studies also fall short in establishing theoretical frameworks that link AI-powered feedback to these behavioral changes, leaving gaps in both empirical research and practical application. This study seeks to address these overlooked dimensions by exploring these behavioral aspects in depth and providing a more comprehensive understanding of AI's role in EFL learning.

Data and methods

Variable measurement. The control variables include age (*Age*), gender (*Gender*), and language proficiency level (*LPL*), i.e., beginner, intermediate, and advanced; and these variables are the demographic description of the respondents. With respect to the first objective, the independent variables include the types of AI-powered feedback, namely corrective AI-powered feedback (*CAIF*), e.g., grammar, vocabulary corrections; and motivational AI-powered feedback (*MAIF*), e.g., encouragement, progress tracking) where each contains three variables: effectiveness of *CAIF* in identifying language errors (*CAIF1*); whether *CAIF* helps in understanding language rules better (*CAIF2*); ability to revise work based on corrective AI-powered feedback (*CAIF3*); effectiveness of *MAIF* for language learning (*MAIF1*); whether motivational AI-powered feedback increase confidence in language learning (*MAIF2*); and ability to continue language learning due to motivational AI-powered feedback (*MAIF3*). However, the dependent variable self-reflection (*SR*) contains three variables, i.e., frequency of reflecting English language learning progress (*SR1*); confident in identifying areas for improvement in English language skills (*SR2*); and frequency of setting goals for improving English language skills (*SR3*).

With respect to the second objective, the independent variables include how helpful is AI-powered feedback in generating new ideas for writing/speaking (*AIFC1*); whether AI-powered feedback encourage to take risks and try new language structures (*AIFC2*); the likeness to use AI-powered feedback to explore different writing/speaking styles (*AIFC3*); and frequency of using AI-powered feedback to brainstorm ideas, to organize thoughts, to evaluate language choices, and to revise and edit (*AIFC4*). However, the dependent variable creativity (*C*) contains three variables, i.e., how often to think creatively in English language learning (*C1*); how confident in expressing ideas in English (*C2*); and whether enjoy writing/speaking in English because it allows to express creativity (*C3*).

With respect to the third objective, the independent variables include whether AI-powered feedback helped reduce anxiety when speaking/writing in English (*AIFRA1*); how helpful is AI-powered feedback in building confidence in English language abilities (*AIFRA2*); whether AI-powered feedback make feel more comfortable taking risks in English language learning (*AIFRA3*); and whether AI-powered feedback changed anxiety level in these situations (*AIFRA4*). While with respect to the mediation variables, include the degree of familiarity with Artificial Intelligence (AI) (*AIFM1*) and degree of preferences to receive feedback on English language learning, such as written comments, audio/video recordings, and face-to-face conversations (*AIFM2*). However, the dependent variable anxiety reduction (*RA*) contains three variables, i.e., how anxious feel when speaking/writing in English (*RA1*); whether worrying about making mistakes in English (*RA2*); and how nervous feel when speaking English (*RA3*).

With respect to the fourth objective, the independent variables include whether using AI-Powered tools improved emotional resilience in English language learning (*AIFER1*); how helpful are AI-Powered tools in managing language learning stress (*AIFER2*); whether AI-Powered tools make feel more motivated to learn English (*AIFER3*); and is there a long-term improvement in emotional resilience since using AI-Powered tools (*AIFER4*). However, the dependent variable emotional resilience (*ER*) contains three variables, i.e., one's degree of emotional resilience in English language learning (*ER1*); whether ones feel overwhelmed by language learning challenges (*ER2*); and how confident one is in handling language learning setbacks (*ER3*).

Instruments/data collection process. This study has conducted a survey to collect data through semi-structured and structured questionnaires aligned with the research questions. The study's target population comprises EFL undergraduate learners across the various Chinese universities. Respondents were selected randomly from the target population; where the sampling method employed is criterion sampling method, which is a variant of the purposive sampling technique used by researchers to select participants who meet particular criteria (Palys, 2008). Considering the distribution of the EFL undergraduate students in the various respective Chinese universities and the required sample size of the application of Structural Equation Modeling¹, with desired margin error of 3% and 95% confidence level, a sample of 205 non-English major EFL undergraduate learners was drawn.

Validity and reliability. To evaluate the validity and reliability of the instrument (Friedman, 2011), a pilot phase was executed with a cohort of English learners analogous to the research participants, where the corrected questionnaire administered randomly on the targeted population and the reliability of the data analyzed through the use of the Cronbach's alpha reliability test which is a method utilized by researchers to confirm that a designed or accepted instrument serves its intended purpose; hence, a test for the validation and reliability of the instruments (Taber, 2018). For each group of questions representing a particular phenomenon focus of the study, Cronbach's alpha reliability test coefficients are between 0.78–0.85, satisfying that the questionnaire is reliable for the data collection (UCLA Statistical Consulting Group, 2020).

Ethical standards. To adhere to the ethical standards set by the research ethics committee, it is important to note that participation in the electronic survey is optional for respondents, ensuring convenience and health safety. Therefore, consent is considered automatic when the respondent agrees to complete the questionnaire. The study will outline the primary objectives in the introduction of the questionnaire and assure respondents of confidentiality, as no personal identifiers such as names or email addresses will be collected during the questionnaire process. The research complied with rigorous ethical standards, implementing procedures for informed consent and data anonymization to protect participant privacy and confidentiality. All study protocols underwent review to confirm that the research was conducted in an ethical and responsible manner.

Functional specifications of the study. This study has four specific objectives to achieve where each is based on a particular dependent variable. Thus, four different equations need to be specified to present the relationship to be estimated. Therefore, the functional specifications specified as in Eqs. 1 through 4 as follows:

$$\begin{aligned} &^{SR_1, SR_2, SR_3} \\ \text{Self - Reflection} &= \beta_0 + \beta_1 \text{age} + \beta_2 \text{gender} + \beta_3 \text{LPL} \\ &+ \beta_4 \text{CAIF}_1 + \beta_5 \text{CAIF}_2 + \beta_6 \text{CAIF}_3 + \beta_7 \text{MAIF}_1 \\ &+ \beta_8 \text{MAIF}_2 + \beta_9 \text{MAIF}_3 + \mu_t \end{aligned} \quad (1)$$

$$\begin{aligned} &^{C_1, C_2, C_3} \\ \text{Creativity} &= \beta_0 + \beta_1 \text{age} + \beta_2 \text{gender} + \beta_3 \text{LPL} + \beta_4 \text{AIFC}_1 \\ &+ \beta_5 \text{AIFC}_2 + \beta_6 \text{AIFC}_3 + \beta_7 \text{AIFC}_4 \\ &+ \beta_8 \text{AIFC}_5 + \mu_t \end{aligned} \quad (2)$$

$$\begin{aligned} &^{AR_1, AR_2, AR_3} \\ \text{Anxiety Reduction} &= \beta_0 + \beta_1 \text{age} + \beta_2 \text{gender} + \beta_3 \text{LPL} \\ &+ \beta_4 \text{AIFAR}_1 + \beta_5 \text{AIFAR}_2 + \beta_6 \text{AIFAR}_3 \\ &+ \beta_7 \text{AIFAR}_4 + \beta_8 \text{AIFM}_1 + \beta_9 \text{AIFM}_2 + \mu_t \end{aligned} \quad (3)$$

$$\begin{aligned}
 ER_1, ER_2, ER_3 \\
 \text{Emotional Resilience} = \beta_0 + \beta_1 \text{age} + \beta_2 \text{gender} + \beta_3 \text{LPL} \\
 + \beta_4 \text{AIFER}_1 + \beta_5 \text{AIFER}_2 + \beta_6 \text{AIFER}_3 \\
 + \beta_7 \text{AIFER}_4 + \mu_t
 \end{aligned}
 \quad (4)$$

where *LPL* means language proficiency level (beginner/intermediate/advanced), *CAIF*₁, *CAIF*₂, and *CAIF*₃ are the questions related to the impact of corrective AI-powered feedback (e.g., grammar, vocabulary corrections) in relation to self-reflection (SR) processes in EFL learners while *MAIF*₁, *MAIF*₂, and *MAIF*₃ are the questions related to the impact of motivational AI-powered feedback (e.g., encouragement, progress tracking) in relation to self-reflection (SR) processes in EFL learners; *AIFC*₁, *AIFC*₂, *AIFC*₃, *AIFC*₄ and *AIFC*₅ are the questions related to the impact of AI-powered feedback in relation to creativity (C) processes in EFL learners; *AIFAR*₁, *AIFAR*₂, *AIFAR*₃, and *AIFAR*₄ are the questions related to the impact of AI-powered feedback in relation to anxiety reduction (AR) processes in EFL learners; *AIFM*₁ and *AIFEM*₂ are the questions related to the mediation impact of AI-powered feedback (learner's familiarity with AI and feedback delivery style) in relation to anxiety reduction (AR) processes in EFL learners; *AIFER*₁, *AIFER*₂, *AIFER*₃, and *AIFER*₄ are the questions related to the impact of AI-powered feedback in relation to emotional resilience (ER) processes in EFL learners. However, $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6, \beta_7, \beta_8$, and β_9 are the coefficients of the respective parameters, while μ_t is the stochastic term of the estimate. Yet, from the equations, the variables age, gender, *LPL*, *CAIF*₁, *CAIF*₂, *CAIF*₃, *MAIF*₁, *MAIF*₂, *MAIF*₃, *AIFC*₁, *AIFC*₂, *AIFC*₃, *AIFC*₄, *AIFC*₅, *AIFAR*₁, *AIFAR*₂, *AIFAR*₃, *AIFAR*₄, *AIFER*₁, *AIFER*₂, *AIFER*₃, and *AIFER*₄ are the independent variables. *AIFM*₁ and *AIFEM*₂ serve as mediating variables in Eq. 3, while *SR*₁, *SR*₂, *SR*₃, *C*₁, *C*₂, *C*₃, *RA*₁, *RA*₂, *RA*₃, *ER*₁, *ER*₂, and *ER*₃ are the dependent variables.

Data analysis techniques. The techniques to be employed include Structural Equation Modeling (SEM) for the quantitative data analysis and phenomenological approach for the qualitative data analysis (specifically the open-ended questions) to explore the influences/acquaintances of AI-powered tools with respect to self-reflection, creativity, and reducing performance anxiety of EFL learners. However, the selection of the SEM technique in this study is attributed to its reputation for evaluating complex interactions involving several dependent and independent variables within a unified analytical framework. This feature makes it an ideal tool for assessing theoretical models that propose causal relationships and interdependencies among variables, as demonstrated in this study (Directory of Statistical Analysis, 2024). Hence, it is due to the issue that in this study under each of the specific objectives of the study, there is multiple dependent variables and independent variables. In addition, under each of the specific objective of this study, each of the dependent variable (the observed variables) has a number of variables that represents it (the latent variables). Moreover, the choice of the phenomenological approach is due to its ability of clear presentation responses in terms of open opinion and its possibility to be estimated using MAXQDA software that relies on the capacity of "Computer assisted qualitative data analysis" programs to enhance the trustworthiness of the qualitative analysis process (Baralt, 2011; Dai and Liu, 2024). The findings of the PA technique are to generate the data for making policy suggestion of the study. However, to ensure the robustness of the SEM findings, this study further employs the PCA technique to construct an index from latent variables, transforming them into an observable

variable similar to a time-series variable. This approach allows the study to apply the Quantile Regression (QR) technique to validate the robustness of the estimates obtained from SEM, thereby enhancing the reliability of the findings. The selection of the QR technique is driven by its ability to handle violations of linearity, heteroscedasticity, and normality assumptions. Unlike traditional regression methods, QR does not impose strict distributional requirements, making it a more flexible and reliable approach for robustness checks (Koenker, 2005). However, the software used in executing this study includes IBM SPSS AMOS version 23, MAXQDA version 24, and EVIEWS version 13.

Results presentation

The analysis of this study began with descriptive statistics to understand the statistical properties of the variables, as shown in Table 1. Following the table, all the 205 respondents filled the questionnaire; thus, no missing values in the data where majority of the respondents are females and the average age is 19 years. However, with respect to the language proficiency level (*LPL*) of the respondents, the majority are intermediate, then followed by beginners, and then advanced.

Result presentation of RQ1: How does the types of AI-powered feedback (eg., corrective vs. motivational) causally affect self-reflection processes in EFL learners?. Figure 2 displays the graphical result presentation of the SEM with respect to model 1. It can be glanced from the figure that all the respective explanatory variables, i.e., *CAIF*, *MAIF*, *Age*, *Gender*, and *LPL* are positively related with the response variable, i.e., *SR* as shown along the arrows running from the explanatory variables to the response variable. Looking at the explanatory variables, there is positive correlation between *CAIF* and *MAIF*, *CAIF* and *Gender*, and *CAIF* and *LPL*. However, negative correlation between *CAIF* and *Age* of 0.23, 0.01, and 0.02, -0.18 , respectively. There is positive correlation between *MAIF* and *LPL*, but negative correlation between *MAIF* and *Age*, *MAIF* and *Gender* of 0.03, -0.12 , and -0.01 , respectively. There is positive correlation between *Age* and *Gender*, *Age* and *LPL*, but negative correlation between *Gender* and *LPL* of 0.16, 0.04, and -0.01 , respectively, as shown along the double-headed arrows running between each pair of the explanatory variables.

Table 2 displays the result of the relationship between the type of AI-powered feedback (corrective and motivational) and self-reflection (SR) of the EFL. From the table, corrective AI-powered feedback (*CAIF*) such as grammar and vocabulary corrections and motivational AI-powered feedback (*MAIF*) such as encouragement and progress tracking are positively related with the SR of the EFL learners, and the impacts are statistically significant at 5% and 10% level, respectively. This has also been reaffirmed by the coefficient of the language proficiency level of the learners (*LPL*), which is both significant and positive. When looking at the statistical healthiness tests of the model, it can be seen that the *p*-value of the likelihood ratio (LR) test statistic is 0.07, the value of the comparative fit index test is 0.96, and the value of the root mean square error is 0.05. The rule is that, for a model to be fit, the LR probability value must be greater than 0.05 (Suh, 2006); acceptable model fit is indicated by a CFI value of 0.90 or greater (Hu and Bentler, 1999); and acceptable model fit is indicated by an RMSEA value of 0.06 or less (Hu and Bentler, 1999). For GFI, AGFI, and TLI, the values are 0.93, 0.90, and 0.91, respectively, where the health values are to be greater than or equal to 0.90, i.e., ≥ 0.90 (Hooper et al. 2008). This means that the estimated model is statistically fit. Therefore, considering the content of the questions used in measuring the self-reflection of the EFL learners, it means that corrective related AI-powered feedback

Table 1 Descriptive statistics.

Variable	Obs.	Mean	Std. Dev.	Min	Max
Panel A: Personal information					
Gender	205	0.3121951	0.4645233	0	1
Male	64				
Female	141				
Age	205	19.00976	2.268692	15	33
LPL	205	1.634146	0.4928955	1	3
Beginners	76				
Intermediate	128				
Advanced	1				
Panel B: Corrective vs. motivational self-reflection AI variables					
CAIF1	205	3.273171	0.7818481	1	5
CAIF2	205	3.390244	0.8597209	1	5
CAIF3	205	3.180488	0.919084	1	5
MAIF1	205	3.209756	0.880008	1	5
MAIF2	205	3.185366	0.972555	1	5
MAIF3	205	3.234146	0.8653487	1	5
Panel C: Self-reflection variables					
SR1	205	2.707317	0.7683667	1	4
SR2	205	3.112195	0.841391	1	5
SR3	205	0.7853659	0.4115734	0	1
Panel D: Creativity AI variables					
AIFC1	205	3.346341	0.9456032	1	5
AIFC2	205	3.082927	0.9640352	1	5
AIFC3	205	3.141463	0.9045028	1	5
AIFC4	205	8.731707	3.944619	1	15
AIFC5	205	2.619512	0.9029417	1	4
Panel E: Creativity variables					
C1	205	2.712195	0.868906	1	4
C2	205	2.77561	0.9173653	1	5
C3	205	3.019512	1.028806	1	5
Panel F: Anxiety reduction AI variables					
AIFRA1	205	3.204878	0.8894676	1	5
AIFRA2	205	3.2	0.8307804	1	5
AIFRA3	205	3.263415	0.906879	1	5
AIFRA4	205	3.209756	0.8910791	1	5
AIFM1	205	3.243902	0.856935	1	5
AIFM2	205	10.61951	5.506824	1	20
Panel G: Anxiety reduction variables					
RA1	205	3.180488	0.919084	1	5
RA2	205	3.268293	1.048444	1	5
RA3	205	3.209756	0.9950379	1	5
Panel H: Emotional resilience AI variables					
AIFER1	205	3.063415	0.8110016	1	5
AIFER2	205	3.282927	0.8031502	1	5
AIFER3	205	3.287805	0.9288922	1	5
AIFER4	205	2.95122	0.8731063	1	5
Panel I: Emotional resilience variables					
ER1	205	2.97561	0.7438574	1	5
ER2	205	3.087805	0.8700336	1	5
ER3	205	3.141463	0.7439216	1	5

Source: Author's computation.

such as grammar and vocabulary corrections and motivational related AI-powered feedback such as encouragement and progress tracking are causing the *EFL* learners to improve their self-reflection, identifying areas for improvement in their English language skills, and setting goals for improving their English language skills. The implications of these findings are significant for both pedagogical practice and the design of AI feedback systems. Corrective feedback, particularly in terms of language mechanics like grammar and vocabulary, provides learners with concrete areas for improvement. Meanwhile, motivational feedback, through encouragement and tracking progress, fosters a sense of achievement and forward momentum. Both types of feedback appear to facilitate a deeper level of self-reflection,

enabling learners to identify their strengths and weaknesses and set meaningful goals for language improvement.

Result presentation of RQ2: What is the causal relationship between the use of AI-powered feedback and the development of creativity in *EFL* learners? Figure 3 shows the graphical result presentation of the SEM with respect to model 2. It can be observed from the figure that the respective explanatory variables, i.e., *AIFC*, *Gender*, and *LPL* are positively related to the response variable, i.e., *C* except *Age*, which is negatively related to it as shown along the arrows running from the explanatory variables to the response variable. In line with the explanatory variables, there is a positive correlation between *AIFC* and *LPL*, *Age* and *Gender*, and *Age* and *LPL*. However, negative correlation between *AIFC* and *Age*, *AIFC* and *Gender*, and *Gender* and *LPL* of 0.4, 0.2, 0.4, -0.2, -0.02, and -0.01, respectively, as shown along the double-headed arrows running between each pair of the explanatory variables.

Table 3 reports the result for examining the causal effect between the use of AI-powered feedback and the development of creativity (*C*) in *EFL* learners. In accordance with the table, creativity-related AI-powered feedback *AIFC* coefficient is positively related with the creativity of the *EFL* learners and the impact is statistically significant at 1% level. This has also been supported by the coefficient of the language proficiency level of the learners (*LPL*) which is also positive. From the part of the diagnostic tests of the model, it is evident that the *p*-value of the LR test statistic is 0.06, the value of the comparative fit index test is 0.93, and the value of the root mean square error is 0.05. Been the LR probability value is greater than 0.05 (Suhr, 2006); values of the CFI, GFI, AGFI, and TLI are greater than 0.90 (Hu and Bentler, 1999; Hooper et al. 2008); and the value of RMSEA is below 0.06 (Hu and Bentler, 1999); the model estimated is statistically healthy. Therefore, looking at the content of the questions used in measuring the creativity of the *EFL* learners, it means that creativity-related AI-powered feedback is making the *EFL* learners to be creative in their English, confidence in expressing their created ideas in English, and enjoying writing/speaking in English to express their creativity. These findings have significant implications for both educators and AI feedback system designers. By integrating creativity-oriented feedback mechanisms, educators can promote not only language proficiency but also creativity in their learners. Encouraging creativity in language learning, through personalized AI-powered feedback, helps learners to feel more confident in their ability to express ideas in English and may inspire more engagement with creative tasks, such as writing and speaking.

Result presentation of RQ3: To what extent does AI-powered feedback reduce performance anxiety in *EFL* learners, and how mediating variables (eg., learner's familiarity with AI, feedback delivery style) influence this relationship? Figure 4 illustrates the graphical result presentation of the SEM with respect to model 3. It is evident from the figure that the respective explanatory variable, i.e., *AIFRA*, is positively related to the response variable, i.e., *RA* while *Age*, *LPL*, *Gender*, *AIFM1*, and *AIFM2* are negatively related with it as shown along the arrows running from the explanatory variables to the response variable. According to the explanatory variables, there is a positive correlation between *AIFRA* and *LPL*, *Age* and *Gender*, and *Age* and *LPL* but a negative correlation between *AIFRA* and *Age*, and *AIFRA* and *Gender*, and *Gender* and *LPL* of 0.05, 0.16, 0.04, -0.10, -0.01, and -0.01, respectively, as shown along the double-headed arrows running between each pair of the explanatory variables.

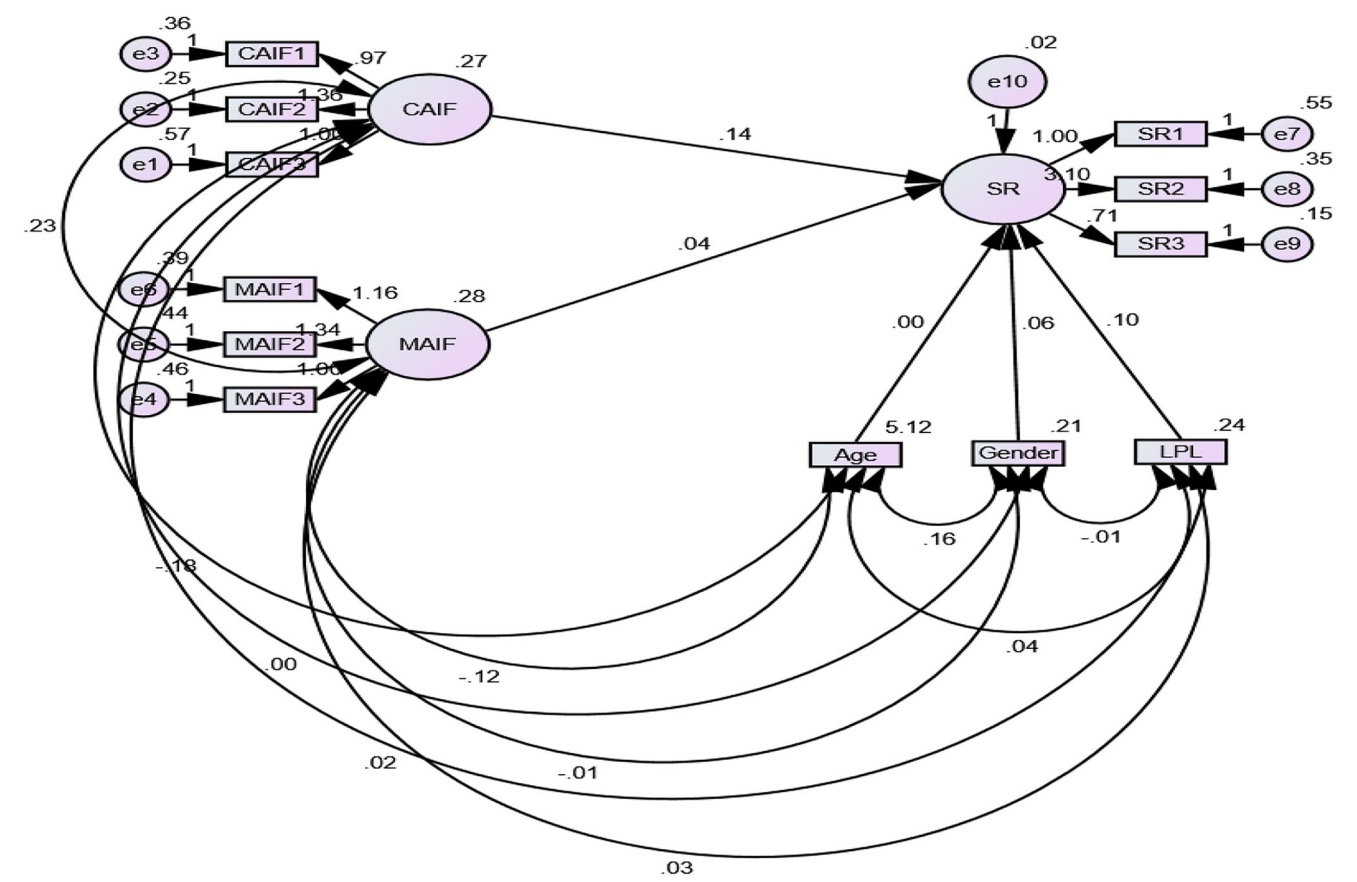


Fig. 2 Result presentation of SEM for model 1.

Table 2 Result of structural equation modeling for model 1.						
Response variable	Direction	Explanatory variables	Estimate	S.E.	C.R.	P
SR	<---	CAIF	0.143	0.123	1.166	0.043**
SR	<---	MAIF	0.042	0.107	0.393	0.094*
SR	<---	Age	0.002	0.008	0.313	0.754
SR	<---	Gender	0.058	0.043	1.354	0.176
SR	<---	LPL	0.104	0.052	1.991	0.046**
Likelihood Ratio (LR) test			p-value	0.07		
Comparative Fit Index (CFI) test			CFI value	0.96		
Root Mean Square Error of Approximation (RMSEA) test			RMSEA value	0.05		
Goodness of Fit Index (GFI)			GFI value	0.93		
Adjusted Goodness of Fit Index (AGFI)			AGFI value	0.90		
Tucker-Lewis Index (TLI)			TLI value	0.91		

Source: Author's computation.
**and * denote significant at 5% and 10 levels of significance, respectively.

Table 4 shows the result for investigating the extent to which AI-powered feedback reduces performance anxiety (RA) in EFL learners, and how mediating variables (e.g., learner's familiarity with AI (AIFM1), feedback delivery style (AIFM2) influence this relationship. From the table, the reduction of anxiety-related AI-powered feedback AIFRA coefficient is statistically insignificant and the sign is not in line with the a priori expectation (i.e., negative); while for the mediation variables, though none is significant, they both satisfied the a priori expectation (i.e., negative) with the anxiety reduction, and this has also been supported by the coefficient of the language proficiency level of the learners (LPL), which is also negative and even statistically significant. When checking for the statistical healthiness of the model, it can be observed that that the *p*-value of the LR test

statistic is 0.07; values of the CFI, GFI, AGFI, and TLI are greater than 0.90 (Hu and Bentler, 1999; Hooper et al. 2008); and the value of the RMSEA is 0.04; the model estimated is statistically healthy. Therefore, with respect to the content of the questions used in measuring the reduction of the performance anxiety of the EFL learners, familiarity of AI-powered feedback and the method in which the learners are receiving feedback on their English learning including by written comments, audio/video recordings, and face-to-face conversations are partially reducing the performance anxiety of the EFL learners when speaking/writing in English, not being worried about making mistakes in English, and not being nervous when you speaking English. These findings suggest that while AI-powered feedback alone may not significantly reduce performance anxiety, the way feedback is

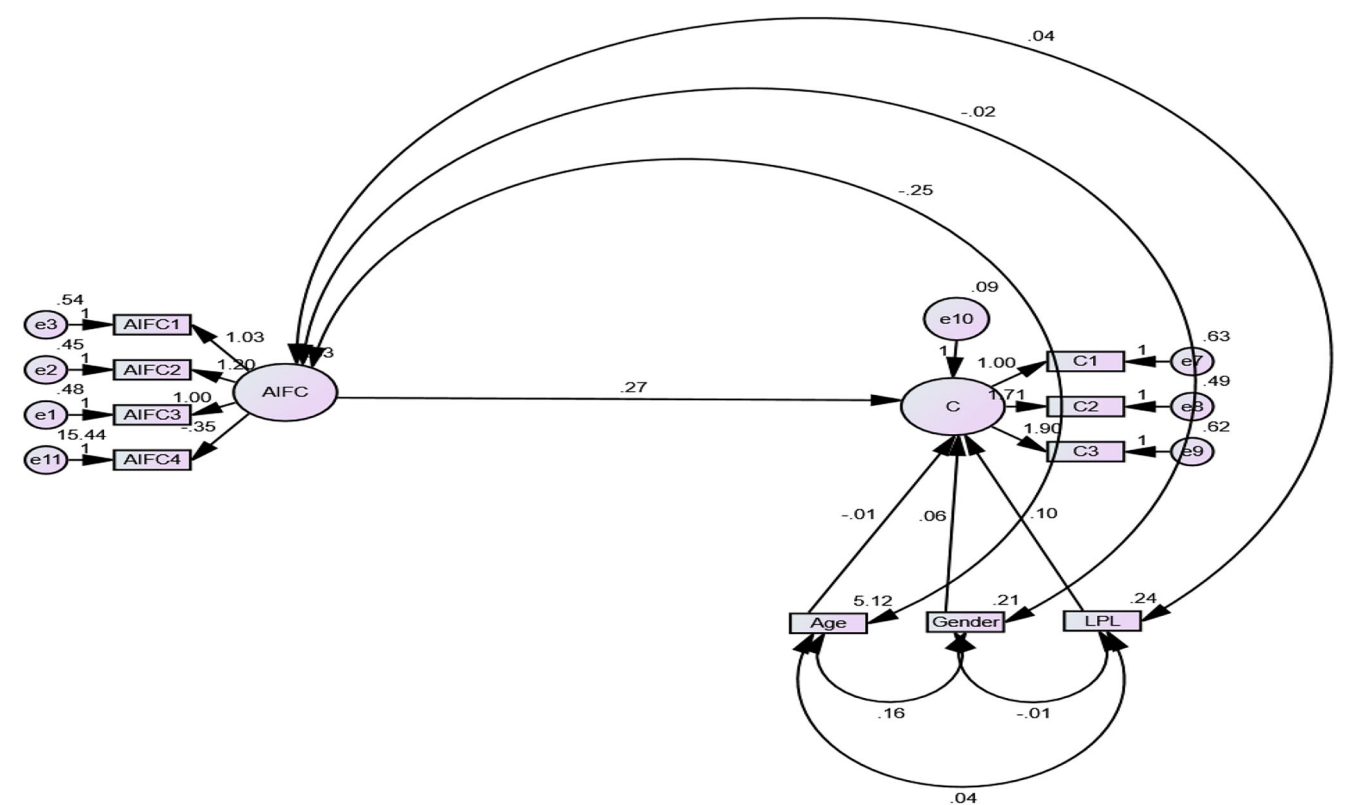


Fig. 3 Result presentation of SEM for model 2.

Table 3 Result of structural equation modeling for model 2.						
Response variable	Direction	Explanatory variables	Estimate	S.E.	C.R.	P
C	<---	AIFC	0.266	0.087	3.038	0.002***
C	<---	Age	−0.013	0.014	−0.920	0.357
C	<---	Gender	0.060	0.065	0.923	0.356
C	<---	LPL	0.099	0.064	1.555	0.120
Likelihood Ratio (LR) test			p-value	0.06		
Comparative Fit Index (CFI) test			CFI value	0.93		
Root Mean Square Error of Approximation (RMSEA) test			RMSEA value	0.05		
Goodness of Fit Index (GFI)			GFI value	0.92		
Adjusted Goodness of Fit Index (AGFI)			AGFI value	0.90		
Tucker-Lewis Index (TLI)			TLI value	0.94		

Source: Author's computation.
***denotes significant at 5% and 10 levels of significance, respectively.

delivered and learners’ familiarity with AI tools could hold the key to alleviating anxiety in EFL contexts. This points to the need for educators and AI developers to focus on not only the content of feedback but also the manner in which it is presented to learners.

Result presentation of RQ4: Does the integration of AI-powered feedback in language learning affect the long-term improvements in EFL learners’ emotional resilience? Figure 5 shows the graphical result presentation of the SEM with respect to model 4. Following the figure, the respective explanatory variable, i.e., *AIFER* and *Age* is positively related to the response variable, i.e., *ER* while *Age* and *LPL* are negatively related to it as shown along the arrows running from the explanatory variables to the response variable. In line with the explanatory variables, there is a positive correlation between *AIFER* and *LPL*, *Age* and *Gender*,

and *Age* and *LPL*, but a negative correlation between *AIFRA* and *Age*, and *AIFRA* and *Gender*, and *Gender* and *LPL* of 0.04, 0.15, 0.04, −0.21, −0.05, and −0.01, respectively, as shown along the double heading arrows running between each pair of the explanatory variables.

Table 5 presents the result for analyzing whether the integration of AI-powered feedback in language learning affects the long-term improvements in EFL learners’ emotional resilience (*ER*). Following the table, the emotional resilience-related AI-powered feedback *AIFER* coefficient is positively related with the emotional resilience of the *EFL* learners and the impact is statistically significant at 1% level. The statistical checks of the model show that the *p*-value of the *LR* test statistic is 0.13; values of the CFI, GFI, AGFI, and TLI are greater than 0.90 (Hu and Bentler, 1999; Hooper et al. 2008); and the value of RMSEA is below 0.06 (Hu and Bentler, 1999); the model estimated is statistically fit. By observing the

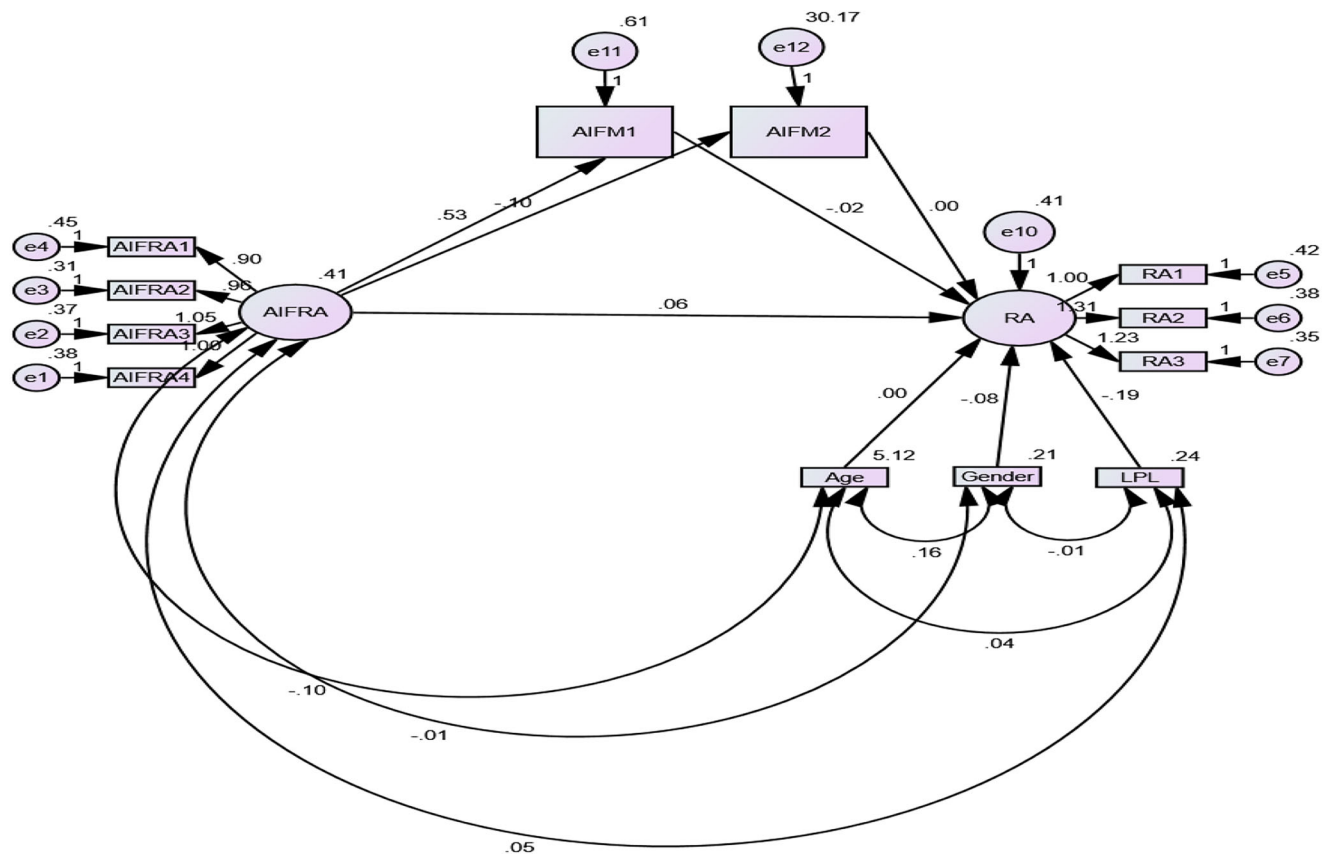


Fig. 4 Result presentation of SEM for model 3.

Table 4 Result of structural equation modeling for model 3.						
Response variable	Direction	Explanatory variables	Estimate	S.E.	C.R.	P
RA	<---	AIFRA	0.063	0.097	0.653	0.514
RA	<---	Age	−0.004	0.022	−0.184	0.854
RA	<---	LPL	−0.193	0.103	−1.869	0.062*
RA	<---	Gender	−0.076	0.108	−0.700	0.484
RA	<---	AIFM1	−0.024	0.065	−0.377	0.706
RA	<---	AIFM2	−0.001	0.009	−0.110	0.912
Likelihood Ratio (LR) test			p-value	0.07		
Comparative Fit Index (CFI) test			CFI value	0.96		
Root Mean Square Error of Approximation (RMSEA) test			RMSEA value	0.05		
Goodness of Fit Index (GFI)			GFI value	0.90		
Adjusted Goodness of Fit Index (AGFI)			AGFI value	0.96		
Tucker-Lewis Index (TLI)			TLI value	0.92		

Source: Author's computation.

*denotes significant at 10% level of significance.

content of the questions used in measuring the emotional resilience of the *EFL* learners, it suggests that emotionally resilient related AI-powered feedback is making the *EFL* learners to be relaxed in English learning challenges and be confident in handling language learning setbacks. These results have significant implications for both educators and the design of AI-powered feedback systems. Emotional resilience is a key factor in sustained language learning success, particularly in environments where learners may face repeated failures or difficulties. AI-powered feedback designed to strengthen emotional resilience could play a crucial role in helping learners stay motivated and maintain a positive outlook, even when confronted with challenging language learning tasks.

Robustness check. Table 6 reports result of quantile regression for model 1 where, according to the table, *CAIF* and *MAIF* are positively related with *SR* and the impact are statistically significant at 5% and 1% levels, and at both 0.25 Qtr. and 0.75 Qtr., respectively. When combining the impacts of the significant coefficients quantiles for each and then compare the level of the impact of the two, that of the *CAIF* is more greater than that of the *MAIF*, thus, corrective related AI-powered feedback is more effective than motivational related AI-powered feedback. Furthermore, from the lower part of the table, it is the quantile slope equality test and the symmetric quantile test. From the quantile slope equality test it is evident that the test rejects the null hypothesis of slope equality at 1% level, which means that the slope equality is different across quantile levels. Likewise, the test

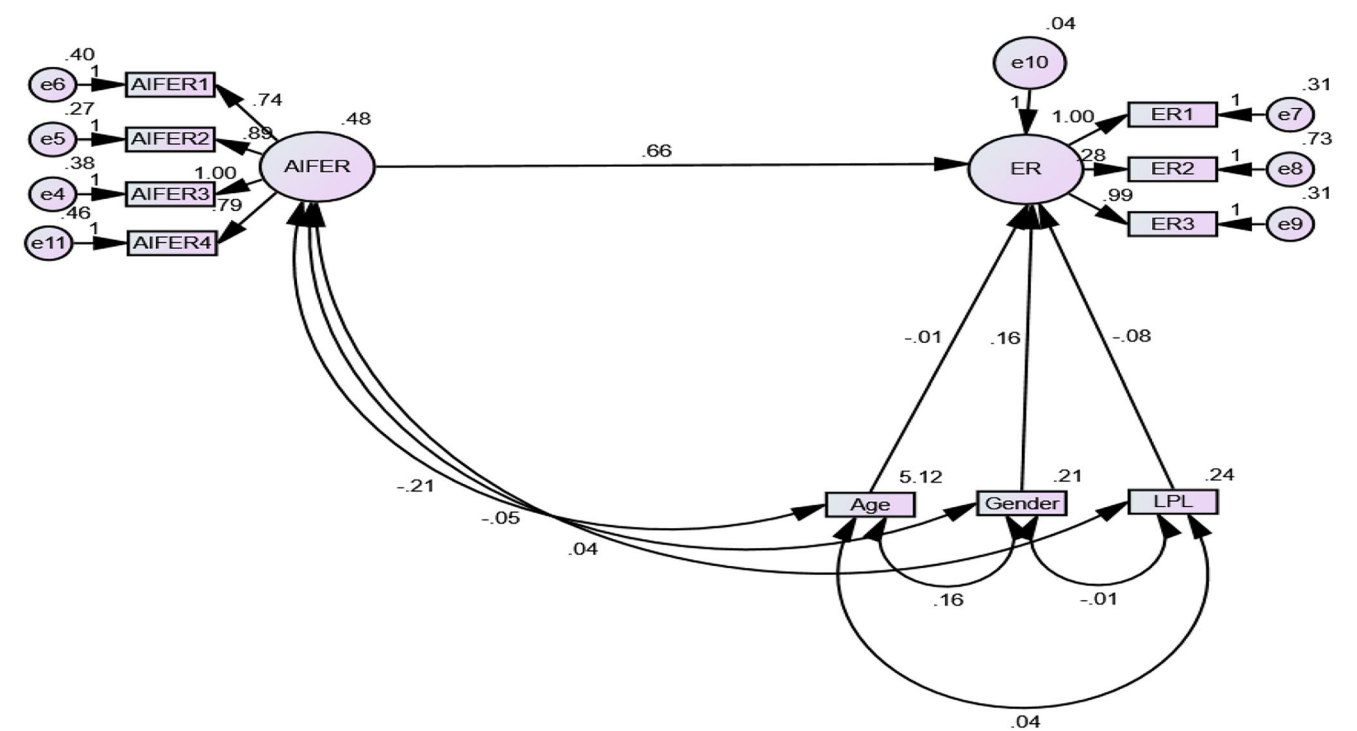


Fig. 5 Result presentation of SEM for model 4.

Table 5 Result of structural equation modeling for model 4.						
Response variable	Direction	Explanatory variables	Estimate	S.E.	C.R.	P
ER	<-	AIFER	0.662	0.086	7.693	***
ER	<---	Age	−0.007	0.016	−0.471	0.638
ER	<---	Gender	0.162	0.078	2.070	0.038**
ER	<---	LPL	−0.080	0.072	−1.118	0.263
Likelihood Ratio (LR) test			p-value	0.13		
Comparative Fit Index (CFI) test			CFI value	0.96		
Root Mean Square Error of Approximation (RMSEA) test			RMSEA value	0.05		
Goodness of Fit Index (GFI)			GFI value	0.94		
Adjusted Goodness of Fit Index (AGFI)			AGFI value	0.93		
Tucker-Lewis Index (TLI)			TLI value	0.95		

Source: Author's computation.
***and ** denote significant at 1% and 5% levels of significance.

Table 6 Quantile regression result for model 1.			
Variable	0.25 Qtr.	0.50 Qtr.	0.75 Qtr.
CAIF	0.413561*	0.019659	0.231922**
MAIF	0.038324	0.003162	0.261038*
AGE	−0.087037*	−0.007186	0.000659
GENDER	0.528987**	0.028762	0.296653***
LPL	0.528983*	0.003610	0.296637
Quantile Slope Equality Test	Chi-square	28.87*	
Symmetric Quantiles Test	Wald Test	36.23*	

Source: Author's computation.
*, **, and *** denote significant at 1%, 5%, and 10% levels of significance, respectively.

of symmetry rejects the null hypothesis of asymmetry at 1% level and thus, there is evidence of asymmetry across the quantiles. Hence, the quantile coefficients across the 0.25, 0.5, and 0.75 are significantly different, and their marginal effects are of significantly different magnitudes. Therefore, CAIF and MAIF are

encouraging the growth of SR. This result of the quantile regression reaffirms the earlier result presented by the SEM estimate for model 1 as presented in Table 2.

Figure 6 is the graphical presentation of coefficients of the quantile regression across the various quantiles for model 2. From the figure, it can be glanced that the variables CAIF and MAIF are significantly and positively influencing SR at lower and upper quantiles, and the impact of the CAIF is greater than that of the MAIF.

Table 7 shows results of quantile regression for model 2. In line with the table, AIFC is positively related with C and the impact is statistically significant at 1% level and at 0.25 Qtr. Furthermore, from the lower part of the table, the quantile slope equality test rejects the null hypothesis of slope equality at 10% level, which suggests that the slope equality is different across quantile levels. Moreover, the test of symmetry rejects the null hypothesis of asymmetry at 5% level, which means evidence of asymmetry across the quantiles. Hence, the quantile coefficients across the 0.25, 0.5, and 0.75 are significantly different; and their marginal effects are significantly of different magnitudes. Thus, AIFC is significantly promoting the growth of SR. Therefore, this result

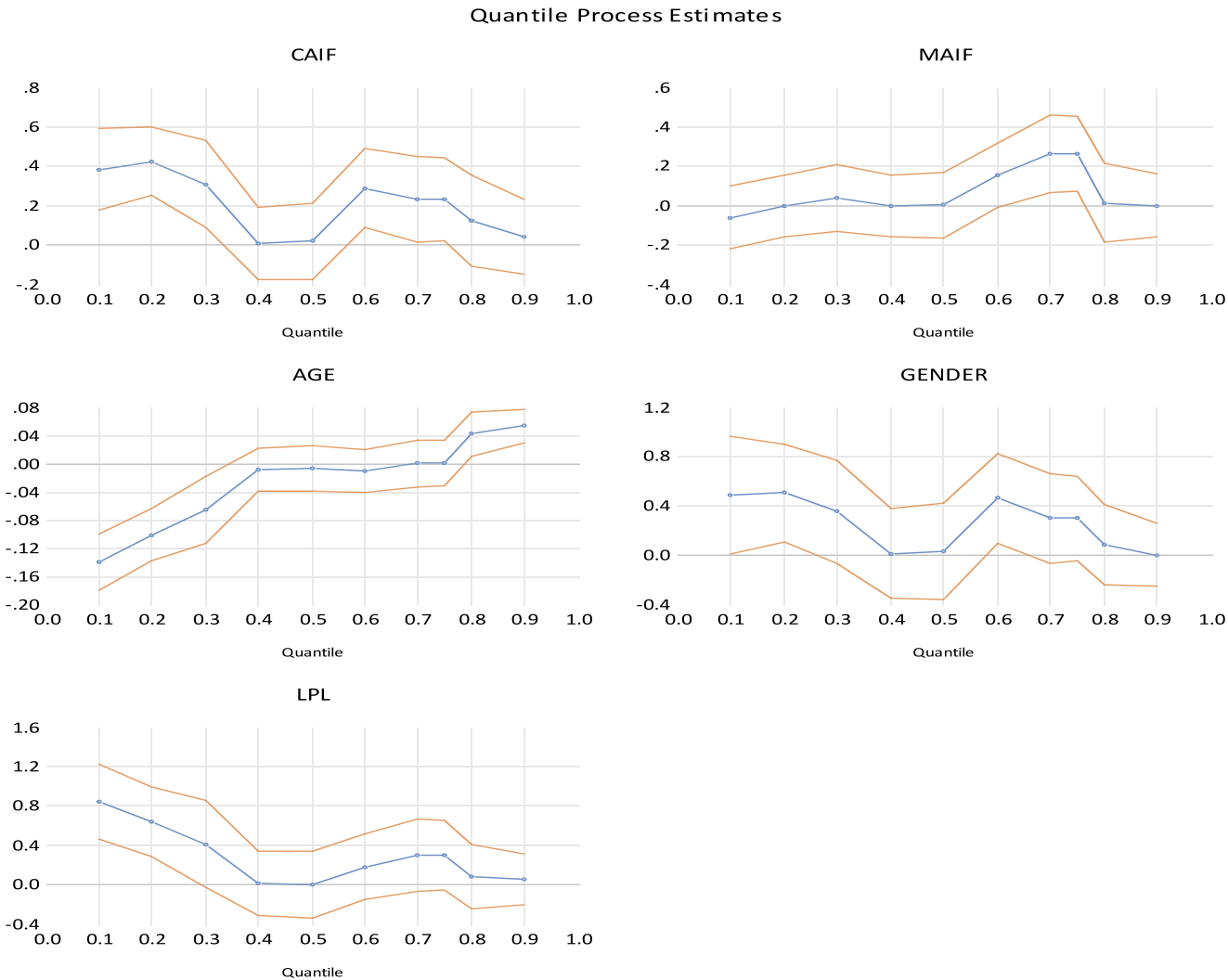


Fig. 6 Graphical presentation of coefficients of the quantile regression for model 1.

Table 7 Quantile regression result for model 2.			
Variable	0.25 Qtr.	0.50 Qtr.	0.75 Qtr.
AIFC	0.226022*	1.82E-05	5.39E-06
AGE	−0.063733*	−0.001053	0.041436**
GENDER	0.493956*	0.001045	0.041429
LPL	0.239017	−9.60E-06	0.082875
Quantile Slope Equality Test	Chi-square	14.67***	
Symmetric Quantiles Test	Wald Test	13.29**	

Source: Author's computation.
*, **, and *** denote significant at 1%, 5%, and 10% levels of significance, respectively.

reiterates the earlier finding reported by the SEM estimate for model 2 as presented in Table 3.

Figure 7 is the graphical presentation of coefficients of the quantile regression across the various quantiles for model 2. From the figure, it can be glanced that the variable *AIFC* is significantly and positively influencing *SR* at lower quantiles.

Table 8 presents result of quantile regression for model 3. In line with the table, *AIFRA* is positively related to *RA* and but the impact is statistically insignificant; however, this violates the a priori, or rather the theoretical assumption of the relationship. Hence, the *AIFRA* is not causing the *RA*. Furthermore, the mediation variables, namely *AIFM1* and *AIFM2* are negatively related with the *RA*, but

neither is significant; however, this is in-line with the a priori or rather the theoretical assumption of the relationship. Moreover, from the lower part of the table, both the quantile slope equality test and the test of symmetry reject the null hypothesis of slope equality and the null hypothesis of asymmetry, both at 1% levels; thus, the slope equality is different across quantile levels, and there is evidence of asymmetry across the quantiles, respectively. Therefore, familiarity with AI-powered feedback (*AIFM1*) and the manner in which EFL learners receive this feedback (*AIFM2*) is partially reducing the performance anxiety of the EFL learners. These findings echo the earlier finding reported by the SEM estimate for model 3 as presented in Table 4.

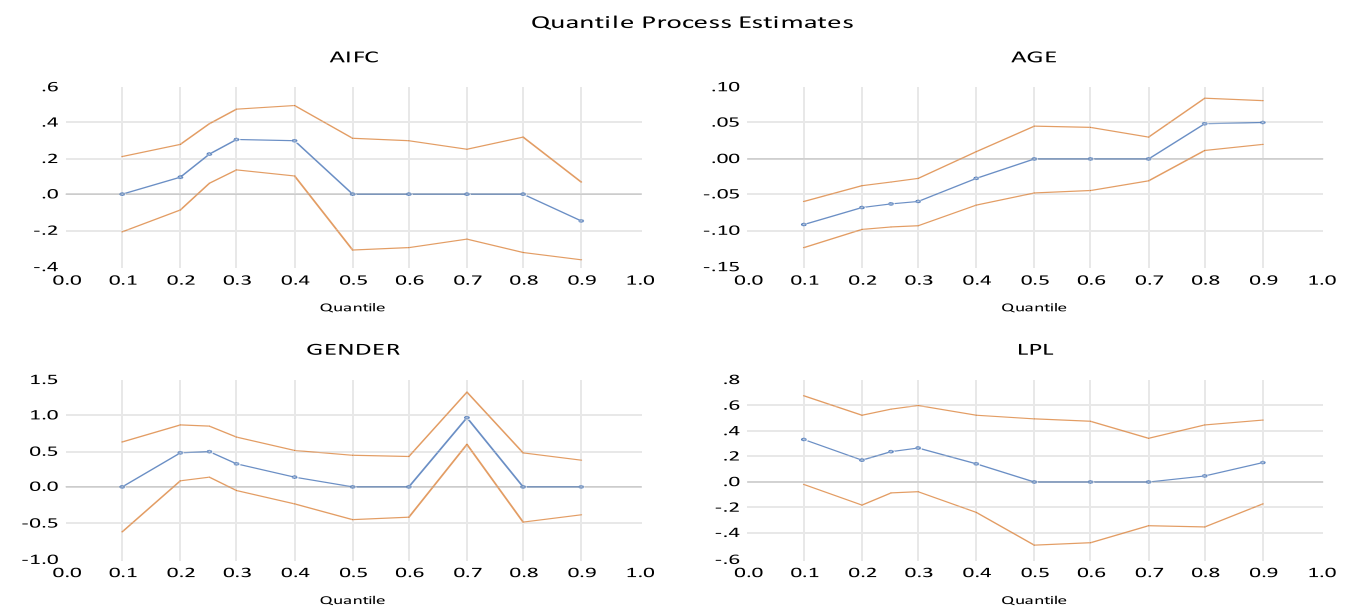


Fig. 7 Graphical presentation of coefficients of the quantile regression for model 2.

Table 8 Quantile regression result for model 3.			
Variable	0.25 Qtr.	0.50 Qtr.	0.75 Qtr.
AIFRA	0.009771	2.49E−19	−0.009600
AIFM1	−0.017259	1.30E−05	0.017874
AIFM2	−0.029790	5.61E−19	0.002021
AGE	−0.010972*	3.93E−18	0.040121*
GENDER	−0.222641	−4.21E−18	−0.029327
LPL	−0.435895*	−0.105424	−0.013806*
Quantile Slope Equality Test	Chi-square	38.23*	
Symmetric Quantiles Test	Wald Test	30.45*	

Source: Author's computation.
*denotes significant at 1% level of significance.

Figure 8 is the graphical presentation of coefficients of the quantile regression across the various quantiles for model 3 which shows that the variable AIFRA is positively related to the RA and the impact is significant, whereas the variables AIFM1 and AIFM2 are positively related to the RA, but neither is significant. Hence, AIFM1 and AIFM2 are partially reducing the RA but AIFRA has no impact on it.

Table 9 indicates result of quantile regression for model 4. In line with the table, AIFER is positively related with RA and the impact is statistically insignificant at 1% and 10% levels, and at 0.25Qtr. and 0.75Qtr., respectively. Furthermore, from the lower part of the table, both the quantile slope equality test and the test of symmetry reject the null hypothesis of slope equality and the null hypothesis of asymmetry both at 1% levels; which means that the slope equality is different across quantile levels and there is evidence of asymmetry across the quantiles, respectively. Therefore, AI-powered feedback related to emotional resilience (AIFER) is enhancing emotional resilience (ER) of the EFL learners. This finding confirms the earlier finding reported by the SEM estimate for model 4 as presented in Table 5.

Figure 9 is the graphical presentation of coefficients of the quantile regression across the various quantiles for model 4, which reveals that the variable AIFER is positively related to the ER, and the impact is significant around lower and upper quantiles.

Discussion

Results presented by this study revealed support that AI-powered feedback is inspirationally promoting the behavioral elements of the learners of EFL, including self-reflection, creativity, reducing performance anxiety, and emotional resilience of the learners. This finding shows that AI tools are worth allowing to be used by the student, though with caution and to some degrees in order not to neglect their natural skills with respect to that. In fact, limits and checks should be put in order while allowing the integration to take place. Moreover, teachers should also understand that AI tools are not something to just be thrown away in teaching and learning of EFL, in fact, a promoter in that matter.

This finding is compatible with the finding of Vera's (2023) perspective that AI presents innovative opportunities to improve language learning experiences and offer personalized support for EFL learners. This perspective is consistent with the findings of Klimova et al. (2023), which present substantial evidence for the beneficial effects of AI on language education. Similarly, Yang (2024) found that AI enhances the personalization of English learning. The results correspond with the conclusions of research conducted by Kim (2019), Schmidt-Fajlik (2023), and Yan (2023), demonstrating that AI-assisted language learning applications markedly enhance the overall performance of English language learners, along with particular language skills and sub-skills. This study's findings corroborate the notion that AI-driven language learning aids facilitate self-regulated learning and augment motivation in English language learners. The claim that AI-assisted instruction, as demonstrated by ChatGPT, significantly improved EFL students' writing skills, motivation, organization, coherence, grammar, and vocabulary compared to traditional instruction. The research by Nugroho et al. (2023) illustrates that AI technologies augment genuine interactions and elevate student productivity and involvement in language learning activities, thereby validating the conclusions of this study. Klayklung et al. (2023) substantiate this study's conclusions concerning AI's ability to provide customized learning experiences and facilitate significant interactions, highlighting its potential in language acquisition. This further emphasizes the importance of AI in linguistics, as evidenced by this study. Furthermore, multiple investigations validate the results of this research, including those conducted by Ali et al. (2023), Al-Obaydi et al. (2023), Kim et al. (2023), Kohnke et al. (2023), and Leunard et al. (2023). These

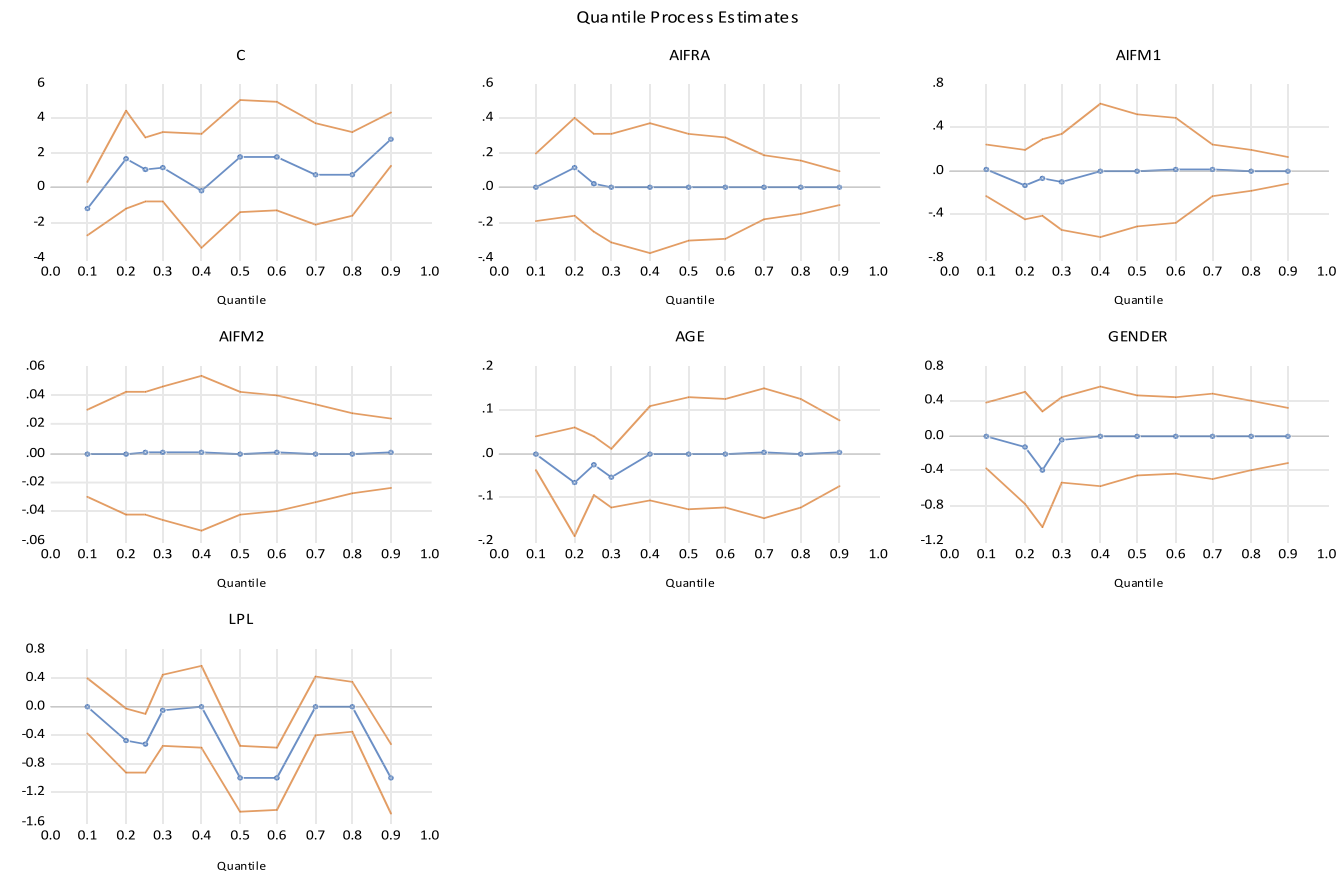


Fig. 8 Graphical presentation of coefficients of the quantile regression for model 3.

Table 9 Quantile regression result for model 4.			
Variable	0.25 Qtr.	0.50 Qtr.	0.75 Qtr.
AIFER	0.530036*	0.001340	0.574668***
AGE	−0.024466	0.001663	0.012381
GENDER	1.80E−07	−0.001655	0.604077**
LPL	−3.22E−05	0.001658	0.012379
Quantile Slope	Chi-square	35.68*	
Equality Test			
Symmetric	Wald Test	19.33*	
Quantiles Test			

Source: Author's computation.
*denotes significant at 1% level of significance.

studies validate the effectiveness of AI technologies in language acquisition and their ability to enhance the language learning experience and revolutionize language instruction. Nonetheless, additional research is necessary to thoroughly evaluate their influence and potential in various learning environments.

The comparison of these findings with existing literature highlights the importance of AI in linguistics, particularly in providing customized learning experiences and facilitating significant interactions. Furthermore, multiple investigations validate the effectiveness of AI technologies in language acquisition and their potential to revolutionize language instruction. Likewise, it is essential to acknowledge the need for further research to thoroughly evaluate the influence and potential of AI-powered feedback in various learning environments. By doing so, educators and policymakers can harness the benefits of AI while ensuring that its integration is done judiciously and with caution.

Conclusion and policy recommendations

This study is an empirical one inspired by AI-powered feedback in enhancing self-reflection, creativity, reducing performance anxiety, and emotional resilience among EFL learners in higher education, with the aim of addressing the growing interest in pedagogical technologies that are AI-driven and their potential to revolutionize language instruction. The sample size is 205 EFL undergraduate learners who are non-English majors from various educational institutions in China, using a purposive criterion sampling technique. The technique used in collecting the data covers a structured and semi-structured questionnaire. The estimation methods employed to conduct the analysis include SEM for the quantitative analysis and the phenomenological technique for the qualitative analysis of the open-ended questions. However, to check for the robustness of the findings, the PCA technique and QR model were used to re-ensure the findings of the study. The results revealed that: (i) corrective-related AI-powered feedback, such as grammar and vocabulary corrections and motivational-related AI-powered feedback, such as encouragement and progress tracking, are causing the EFL learners improving their self-reflection, identifying areas for improvement in their English skills, and setting goals for improving their English skills. However, when comparing the level of the impact of the two, corrective related AI-powered feedback is more effective than the motivational related AI-powered feedback (ii) creativity related AI-powered feedback is making the EFL learners to be creative in their English, confidence in expressing created ideas in English, and enjoying writing/speaking in English to express their creativity (iii) familiarity of AI-powered feedback and the method in which the EFL learners are receiving feedback of their English learning are partially reducing the learners' performance anxiety (iv)

Quantile Process Estimates

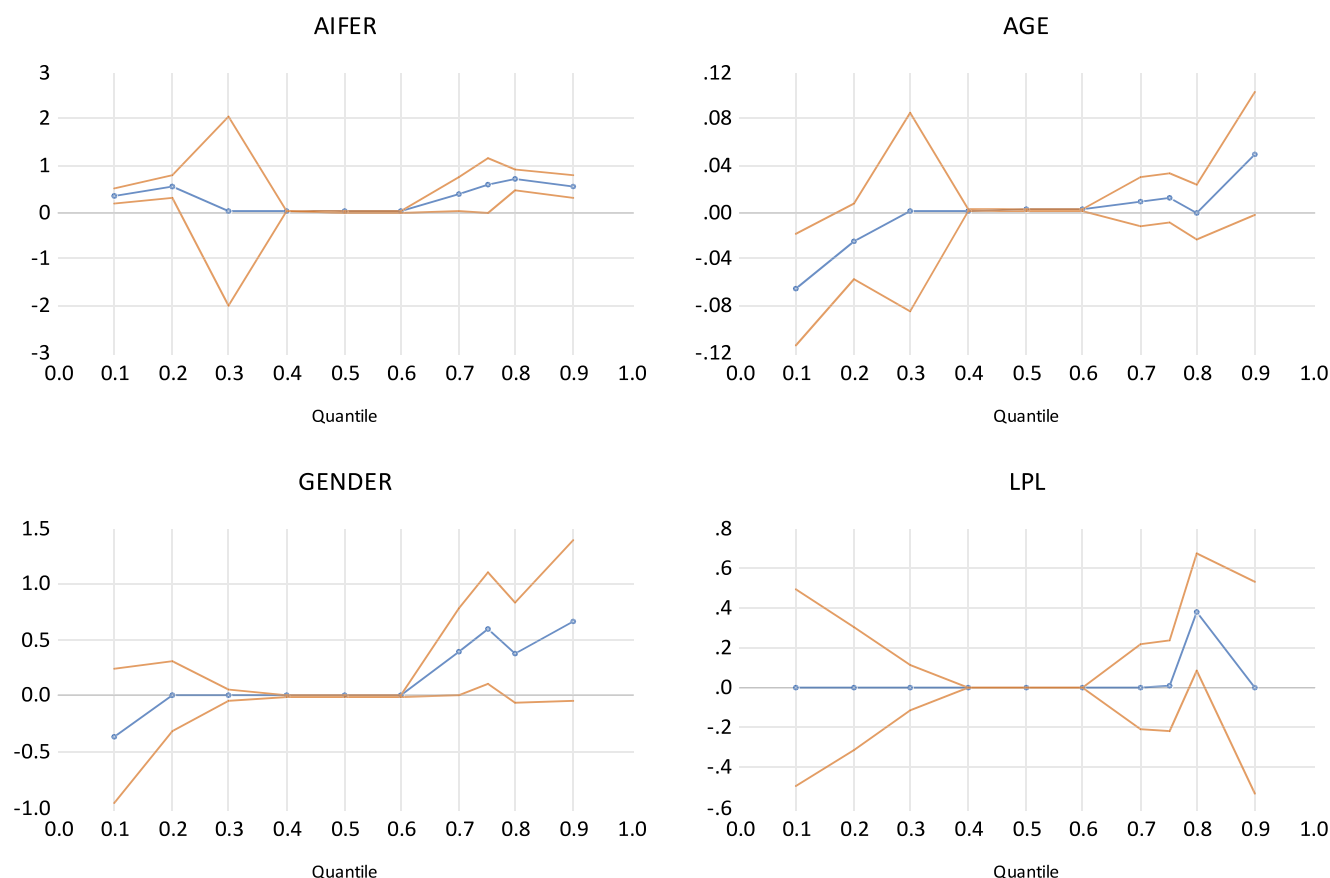


Fig. 9 Graphical presentation of coefficients of the quantile regression for model 4.

emotional resilience related AI-powered feedback is relaxing the *EFL* learners English learning challenges and being confident in handling language learning setbacks.

Policy recommendations to AI vendors/companies. Considering the analysis of the open-ended questions of the study regarding how AI-powered feedback could be improved to enhance self-reflection in language learning, support creativity in language learning, reduce performance anxiety, and support emotional resilience, this study offers the following recommendations to the AI vendors/companies for improvement:

- i. On how to improve AI-powered feedback for enhancing self-reflection in *EFL* learners:

AI-powered feedback could be improved to enhance self-reflection by enhancing the AI model and training data to improve accuracy, implement error tracking and logging functions to review and address errors, provide more real-life examples in the AI-powered feedback to increase user engagement, offer personalized tutorials and guidance to users, improve the AI's intelligence and analytical capabilities, incorporate self-reflection and self-evaluation elements, focus on enhancing human-AI interaction and conversation to improve user experience, add more emotional intelligence and empathy to the AI's feedback and responses, and increase the precision and accuracy of the AI's error correction and suggestions.

- ii. On how to improve AI-powered feedback for developing creativity in *EFL* learners:

AI-powered feedback could be improved to develop the creativity in *EFL* learners by enhancing the interaction and evaluation capabilities of AI to generate diverse ideas and perspectives beyond superficial descriptions, improve AI's language understanding and generation to be more aligned with human habits, using more natural and contextual language, provide personalized feedback and suggestions tailored to learners' interests and learning styles to encourage creativity and experimentation in language use, expand AI's knowledge base and reasoning abilities to offer more accurate, practical, and intellectually engaging responses, going beyond simple answers, incorporate elements of gamification, peer feedback, and open-ended activities to foster a stimulating environment for creative language learning, strengthen AI's comprehension of cultural nuances, idiomatic expressions, and intended tone/emotion to provide more contextually appropriate and insightful feedback, enhance the AI's ideological latitude and ability to generate a wider range of vocabulary and expressions relevant to different domains and industries.

- iii. On how to improve AI-powered feedback for reducing performance anxiety in *EFL* learners:

AI-powered feedback could be improved in reducing performance anxiety in *EFL* learners by personalize recommendations and provide real-time emotional support, adapt to different occasions and styles of expression needs, use interesting visuals and words of encouragement to humanize the experience, increase conversational communication and interaction, incorporate methods related to anxiety relief and stress management, collect more insights

about real people's thinking to improve intelligence and personalization, provide motivational and empathetic feedback, focusing on positive reinforcement and progress, adopt a gentle, supportive, and constructive approach to reduce performance anxiety, incorporate elements of cognitive-behavioral therapy to help users reframe negative thought patterns, and include stress-detection features to prompt the system to provide calming strategies or motivational content in real-time.

- iv. On how to improve AI-powered feedback for emotional resilience in EFL learners:

To improve AI-powered feedback for the emotional resilience in EFL learners, the AI-powered tools could offer more personalized and timely interventions, such as customized coping strategies or mindfulness exercises, triggered by detecting signs of emotional distress. Integrating more human-like empathy and a greater variety of emotional support resources would also enhance their ability to bolster emotional resilience effectively.

Policy recommendations to stakeholders. Considering the findings from the quantitative analysis of the study, the following recommendations were provided to authorities on how to with righteous arrangement propagate the use of AI-powered feedback tools in the teaching and learning process of EFL.

- i. On how authorities should improve AI-powered feedback for enhancing the self-reflection of the EFL learners: Authorities should incorporate AI-driven feedback instruments into the national EFL curriculum to enhance personalized learning and self-assessment; create continuous AI-focused professional development programs for EFL educators; and establish standards for ethical usage of AI-driven feedback tools, ensuring data confidentiality and student consent.
- ii. On how to improve AI-powered feedback for the development of creativity of the EFL learners: There is need to incorporate AI-driven creative writing tools into the EFL curriculum to enhance students' imagination and creativity; establish collaborative initiatives that leverage AI-driven feedback to augment creativity and cooperation among EFL learners; and facilitate continuous professional development for educators to proficiently incorporate AI tools that enhance creativity in EFL classroom.
- iii. On how to improve AI-powered feedback for reducing performance anxiety of the EFL: The stakeholders should embed AI-driven formative assessments inside the EFL curriculum to facilitate ongoing, low-stakes feedback; utilize AI feedback to cultivate a growth mindset in EFL learners by prioritizing effort and progress over inherent talent; and formulate explicit protocols for the ethical application of AI in education, emphasizing data privacy and student permission.
- iv. On how to improve AI-powered feedback for emotional resilience in EFL learners:

Stakeholders should embed AI-powered emotional support tools inside the EFL curriculum to deliver immediate, tailored feedback that caters to students' emotional requirements; utilize AI feedback to cultivate a constructive and helpful educational atmosphere that promotes emotional resilience; create explicit ethical guidelines and data privacy policies governing the application of AI in education, emphasizing the safeguarding of students' emotional welfare.

Limitation/suggestion for future studies. Despite the fact that the research questions that this study answers are novel, this

study has a limitation. The limitations of this study include the fact that it investigates the inspiration of AI-powered feedback in enhancing self-reflection, creativity, and reducing performance anxiety among learners of EFL from the perspective of learners only; consequently, it did not evaluate teachers' perspectives on such an issue. In this regard, it is suggested that future research on AI-powered feedback in relation to such behavioral factors should be conducted on teachers to overcome this deficiency. In addition, the study relies on self-reported responses from structured and semi-structured questionnaires, which may introduce subjective bias and social desirability effects in how learners perceive AI-powered feedback. A mixed-method approach incorporating longitudinal observations and experimental designs could strengthen reliability. Furthermore, while the study assesses short-term effects, it does not explore long-term impacts of AI-powered feedback on learners' self-reflection, creativity, anxiety reduction, and emotional resilience. Future studies should conduct longitudinal research to evaluate sustained improvements. Moreover, factors such as teaching styles, prior AI exposure, institutional support, and social influences were not deeply explored in the study. Future research should integrate external mediating factors to provide a more comprehensive perspective on AI-enhanced language learning.

Data availability

The data used and materials are available on request.

Received: 17 March 2025; Accepted: 21 August 2025;

Published online: 29 September 2025

Note

- 1 First, according to the SEM method, the minimum number should be 200 even though some scholars are arguing that the minimum should be 100; and second, it has been stated that the sample size of SEM should be in a ratio of the sample size to the number of parameters as 5:1 up to 20:1, i.e., sample size relative to the number of parameters of the model to be estimated (for these sample size criteria, see, Huber, 2014); however, in this study no model with more than 9 parameters. Therefore, the sample size of this study (205) satisfies the sample criteria of the SEM technique employed to conduct the analysis of the study.

References

- Akpur U (2020) Critical, reflective, creative thinking and their reflections on academic achievement. *Think Skills Creativity* 37:100683. <https://doi.org/10.1016/j.tsc.2020.100683>
- Ali JKM, Shamsan MAA, Hezam TA, Mohammed AA (2023) Impact of ChatGPT on learning motivation: teachers and students' voices. *J Engl Stud Arabia Felix* 2(1):41–49. 1056540/jesaf.v2i1.51
- Al-Obaydi LH, Pikhart M, Klímová B (2023) ChatGPT and the general concepts of education: Can Artificial Intelligence-Driven Chatbots support the process of language learning? *Int J Emerg Technol Learn (IJET)* 18(21):39–50. <https://doi.org/10.3991/ijet.v18i21.42593>
- AlTwijri L, Alghizzi TM (2024) Investigating the integration of artificial intelligence in English as foreign language classes for enhancing learners' affective factors: A systematic review. *Heliyon* 10:e31053
- Alzahrani FKJ, Alotaibi HH (2024) The Impact of Artificial Intelligence on Enhancing EFL Writing Skills among High School Students. <https://doi.org/10.33193/JEAHS.34.2024.482>
- Amabile TM (1996) Creativity in context: Update to "The Social Psychology of Creativity". Westview Press, <https://psycnet.apa.org/record/1996-97996-000>
- Amin MYM (2023) AI and chat GPT in language teaching: Enhancing EFL classroom support and transforming assessment techniques. *Int J High Educ Pedag* 4(4):1–15
- Bakare OD, Jatto OV (2023) The potential impact of Chatbots on student engagement and learning outcomes. In *Creative AI tools and ethical implications in teaching and learning*, 212–229. IGI Global
- Baralt M (2011) Coding qualitative data. Research methods in second language acquisition: A practical guide, 222–244. <https://doi.org/10.1002/9781444347340>
- Beghetto RA (2007) Does creativity have a place in classroom discussions? Prospective teachers' response preferences. *Think skills creativity* 2(1):1–9

- Brown HD (2000) Principles of Language Learning and Teaching, vol. 4, Longman, New York. <https://gustavorubinoernesto.com/wp-content/uploads/2020/06/H-Douglas-Brown-Principles-of-Language-Learning-and-Teaching.pdf>
- Carter R (2015) Language and creativity: The art of common talk. Routledge. <https://doi.org/10.4324/9781315658971>
- Chen X, Zou D, Xie H, Cheng G, Liu C (2022) Two decades of artificial intelligence in education. *Educ Technol Soc* 25(1):28–47. <https://doi.org/10.1007/s10639-022-11209-y>
- Crystal D (2011) Internet linguistics: A student guide. Routledge
- Csikszentmihalyi M (1990) Flow: The psychology of optimal experience. Harper Row. <https://www.scrip.org/reference/referencespapers?referenceid=2313227>
- Dai K, Liu Q (2024) Leveraging Artificial Intelligence (AI) in English as a Foreign Language (EFL) Classes: Challenges and Opportunities in the Spotlight. *Computers in Human Behavior*, 108354. <https://doi.org/10.1016/j.chb.2024.108354>
- Dewaele JM, Chen X, Padilla AM, Lake J (2019) The flowering of positive psychology in foreign language teaching and acquisition research. *Front Psychol* 10:2128
- Dewaele JM, MacIntyre PD (2016) Foreign language enjoyment and foreign language classroom anxiety: The right and left feet of the language learner. *Positive Psychology in SLA*, 215(236), 9781783095360-010
- Directory of statistical analysis (2024) A Comprehensive Guide to Structural Equation Modeling. Retrieved from: <https://www.statisticssolutions.com/free-resources/directory-of-statistical-analyses/structural-equation-modeling/>
- Dishon N, Oldmeadow JA, Critchley C, Kaufman J (2017) The effect of trait self-awareness, self-reflection, and perceptions of choice meaningfulness on indicators of social identity within a decision-making context. *Front Psychol* 8:2034. <https://doi.org/10.3389/fpsyg.2017.02034>
- Erdogan F (2019) Effect of cooperative learning supported by reflective thinking activities on students' critical thinking skills. *Eurasia J Educ Res* 19(80):89–112
- Fan J, Zhang Q (2024) From literacy to learning: The sequential mediation of attitudes and enjoyment in AI-assisted EFL education. *Heliyon*, 10(17)
- Fang T, Yang S, Lan K, Wong DF, Hu J, Chao LS, Zhang Y (2023) Is chatgpt a highly fluent grammatical error correction system? a comprehensive evaluation. *arXiv preprint arXiv:2304.01746*. <https://doi.org/10.48550/arXiv.2304.01746>
- Fathi J, Rahimi M (2022) Examining the impact of flipped classroom on writing complexity, accuracy, and fluency: A case of EFL students. *Comput Assist Lang Learn* 35(7):1668–1706. <https://doi.org/10.1080/09588221.2020.1825097>
- Fitria TN (2023) Artificial intelligence (AI) technology in OpenAI ChatGPT application: A review of ChatGPT in writing English essay. *ELT Forum: J Engl Lang Teach* 12(No. 1):44–58. <https://doi.org/10.15294/elt.v12i1.64069>. In
- Friedman DA (2011) How to collect and analyze qualitative data. *Research methods in second language acquisition: A practical guide*, 180–200
- Gao Y, Wang Q, Wang X (2024) Exploring EFL University Teachers' Beliefs in Integrating ChatGPT and other Large Language Models in Language Education: A Study in China. *Asia Pac J Educ* 44(1):29–44. <https://doi.org/10.1080/02188791.2024.2305173>
- Gkonou C, Daubney M, Dewaele JM (Eds.) (2017) New insights into language anxiety: Theory, research and educational implications (Vol. 114). *Multilingual Matters*
- Guo K, Pan M, Li Y, Lai C (2024) Effects of an AI-supported approach to peer feedback on university EFL students' feedback quality and writing ability. *Internet High Educ* 63:100962. <https://doi.org/10.1016/j.iheduc.2024.100962>
- Hamad Al-Rashidi A, Aberash A (2024) Reflective thinking and self-evaluation in language learning: mirroring the impacts on Saudi Arabian EFL students' growth mindfulness, resilience, and academic well-being. *Asian-Pac J Second Foreign Lang Educ* 9(1):44
- Hooper D, Coughlan J, Mullen MR (2008) Structural Equation Modelling: Guidelines for Determining Model Fit. *Electron J Bus Res Methods* 6(1):53–60
- Horwitz E (2001) Language anxiety and achievement. *Annu Rev Appl Linguist* 21:112–126. <https://doi.org/10.1017/S0267190501000071>
- Hu LT, Bentler PM (1999) Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Struct Equ modeling: a Multidiscip J* 6(1):1–55. <https://doi.org/10.1080/10705519909540118>
- Huang J, Tan M (2023) The role of ChatGPT in scientific communication: writing better scientific review articles. *Am J cancer Res* 13(4):1148
- Huber C (2014) Introduction to Structural Equation Modelling Using Stata. A Paper Presentation at University of Southampton California, Rossier School of Education on 3rd 2016. https://youtu.be/bqTn9fgxotw?si=EKiq_WMvzE0Dsyk
- Hwang WY, Nurtantyan R, Purba SWD, Hariyanti U, Indrihapsari Y, Surjono HD (2023) AI and recognition technologies to facilitate English as foreign language writing for supporting personalization and contextualization in authentic contexts. *J Educ Comput Res* 61(5):1008–1035
- Ilkka T (2018) The impact of artificial intelligence on learning, teaching, and education. *European Union*
- Jafari Z, Yavari S, Shokri M (2015) The impact of self-assessment of EFL learners' tolerance of ambiguity. *Int J Lang Linguist* 3(4):271–274. <https://doi.org/10.11648/j.ijll.20150304.22>
- Johnson S (2010) Where good ideas come from. Allen Lane/Penguin. <https://www.amazon.com/Where-Good-Ideas-Come-Innovation/dp/1594485380>
- Jones RH (2016) Digital literacies. In *Handbook of research in second language teaching and learning* (pp. 286–298). Routledge. <https://centaur.reading.ac.uk/68978/1/Jones%202016%20Digital%20Literacies%20in%20Hinkle%20ed.pdf>
- Jones RH, Richards JC (Eds.). (2015) Creativity in language teaching: Perspectives from research and practice. Routledge. https://www.routledge.com/Creativity-in-Language-Teaching-Perspectives-from-Research-and-Practice/Jones-Richards/p/book/9781138843653?srsltid=AfmBOorziLVvrlAK9LonZtd2CUzDXakBTRHDQJbY_T4u5qzusCuxnhN
- Karimi N, Babaii E, Daftarifard P (2022) The effectiveness of reflective learning through semi-structured reflection writing in reading performance and metacognitive awareness of Iranian EFL learners. *J Lang Transl* 12(1):202–222
- Kim S, Shim J, Shim J (2023) A study on the utilization of OpenAI ChatGPT as a second language learning tool. *J Multimed Inf Syst* 10(1):79–88. <https://doi.org/10.33851/jmis.2023.10.1.79>
- Kim SG (2023) Using ChatGPT for language editing in scientific articles. *Maxillofac Plast Reconstr Surg* 45(1):13. <https://doi.org/10.1186/s40902-023-00381-x>
- Kim NY (2019) A study on the use of artificial intelligence chatbots for improving English grammar skills. *J Digit Conver*, 17(8)
- Klayklung P, Chocksathaporn P, Limna P, Kraiwanit T, Jangjarat K (2023) Revolutionizing education with ChatGPT: Enhancing learning through conversational AI. *Univers J Educ Res* 2(3):217–225. <https://doi.org/10.5281/zenodo.8352965>
- Klimova B, Pikhart M, Benites AD, Lehr C, Sanchez-Stockhammer C (2023) Neural machine translation in foreign language teaching and learning: a systematic review. *Educ Inf Technol* 28(1):663–682
- Koenker R (2005) Quantile regression. Cambridge Univ Pr
- Kohnke L, Moorhouse BL, Zou D (2023) ChatGPT for language teaching and learning. *RELJ* 54(2):537–550. <https://doi.org/10.1177/00336882231162868>
- Kukulska-Hulme A, Viberg O (2018) Mobile collaborative language learning: State of the art. *Br J Educ Technol* 49(2):207–218
- Lei X, Fathi J, Noorbakhsh S, Rahimi M (2022) The impact of mobile-assisted language learning on English as a foreign language learners' vocabulary learning attitudes and self-regulatory capacity. *Front Psychol* 13:872922. <https://doi.org/10.3389/fpsyg.2022.872922>
- Leonard H, Rachmawati R, Zani B, Maharjan K (2023) GPT Chat: Opportunities and challenges in the learning process of arabic language in higher education. *J Int Ling Technol* 2(1):10–22. <https://doi.org/10.55849/jiltech.v2i1.439>
- Liu H, Jin L, Han X, Wang H (2024) Unraveling the relationship between English learning burnout and academic achievement: the mediating role of English Learning resilience. *Behav Sci* 14(12):1124
- Liu H, Fan J (2024) AI-Mediated Communication in EFL Classrooms: The Role of Technical and Pedagogical Stimuli and the Mediating Effects of AI Literacy and Enjoyment. *Eur J Educ*, e12813. <https://doi.org/10.1111/ejed.12813>
- MacIntyre P, Gregersen T (2012) Emotions that facilitate language learning: The positive-broadening power of the imagination. *Stud second Lang Learn Teach* 2(2):193–213
- MacIntyre PD, Gardner RC (1994) The subtle effects of language anxiety on cognitive processing in the second language. *Lang Learn* 44(2):283–305
- Maley A, Kiss T (2017) Creativity and English language teaching: From inspiration to implementation. Springer. <https://link.springer.com/book/10.1057/978-1-137-46729-4>
- Marvell A, Simm D (2021) Encountering emotions during international fieldwork: using innovative pedagogies to develop emotional intelligence and resilience. *Experiential learning in geography: Experience, evaluation and encounters*, 57–75
- Marzuki W, Rusdin U, Darwin D, Indrawati I (2023) The impact of AI writing tools on the content and organization of students' writing: EFL teachers' perspective. *Cogent Educ* 10(2):2236469. <https://doi.org/10.1080/2331186X.2023.2236469>
- Mohebbi H, Coombe C (2021) Research Questions in Language Education and Applied Linguistics. Springer. file:///C:/Users/babar/Downloads/table-ofcontents-RQs-MohebbiandCoombe.pdf
- Mumford S, Dikilitas K (2020) Pre-service language teachers reflection development through online interaction in a hybrid learning course. *Comput Educ* 144:103706
- Nowak A, Vallacher RR, Bartkowski W, Olson L (2022) Integration and expression: The complementary functions of self-reflection. *J Personal* 91(4):947–962. <https://doi.org/10.1111/jopy.12730>
- Nugroho A, Putro NHPS, Syamsi K (2023) The potentials of ChatGPT for language learning: Unpacking its benefits and limitations. *Register J* 16(2):224–247. <https://doi.org/10.18326/register.v16i2.224-247>
- Orakçı Ş (2021) Teachers' reflection and level of reflective thinking on the different dimensions of their teaching practice. *Int J Mod Educ Stud* 5(1):118–149. <https://doi.org/10.51383/ijonmes.2021.88>
- Oxford RL (2016) Teaching and researching language learning strategies: Self-regulation in context. Routledge
- Palys T (2008) Purposive sampling. In L. M. Given (Ed.), *the sage encyclopedia of qualitative research methods* (pp. 697–698). Sage Publications

- Qi S, Derakhshan A (2024) Technology-based collaborative learning: EFL learners' social regulation and modifications in their academic emotions and academic performance. *Education and Information Technologies*, 1–26
- Rhodes M (1961) An analysis of creativity. *Phi Delta Kappan* 42(7):305–310. <https://www.jstor.org/stable/20342603>
- Sarani A, Alibakhshi G, Molazehi H (2014) On application of critical pedagogy principles by ELT instructors and subject teachers: A Case of Iranian universities. *J Res Applied Linguistics* 5(2):41–56
- Sajja R, Sermet Y, Cikmaz M, Cwierny D, Demir I (2024) Artificial intelligence-enabled intelligent assistant for personalized and adaptive learning in higher education. *Information* 15(10):596
- Sawyer RK, Henriksen D (2024) *Explaining creativity: The science of human innovation*. Oxford University Press
- Schmidt-Fajlik R (2023) ChatGPT as a grammar checker for Japanese English language learners: A comparison with Grammarly and ProWritingAid. *AsiaCALL Online J* 14(1):105–119. <https://doi.org/10.54855/acoj.231417>
- Scott G, Leritz LE, Mumford MD (2004) The effectiveness of creativity training: A quantitative review. *Creat Res J* 16(4):361–388
- Selim ASM (2024) The transformative impact of AI-powered tools on academic writing: Perspectives of EFL university students. *Int J Engl Linguist* 14(1):14. <https://doi.org/10.5539/ijel.v14n1p14>
- Shen C, Shi P, Guo J, Xu S, Tian J (2023) From process to product: writing engagement and performance of EFL learners under computer-generated feedback instruction. *Front Psychol* 14:1258286
- Su PH, Wu CH, Lee LS (2014) A recursive dialogue game for personalized computer-aided pronunciation training. *IEEE/ACM Trans Audio, Speech, Lang Process* 23(1):127–141
- Su Y, Lin Y, Lai C (2023) Collaborating with ChatGPT in argumentative writing classrooms. *Assess Writ* 57:100752. <https://doi.org/10.1016/j.asw.2023.100752>
- Suhr D (2006) The basics of structural equation modeling. *Presented: Irvine, CA, SAS User Group of the Western Region of the United States (WUSS)*, 1–19. Retrieved from: <https://lexjansen.com/wuss/2006/tutorials/TUT-Suhr.pdf>
- Syifaudinn M, Yuliansyah AR (2023) The effect of using AI on Students' motivation and anxiety in learning English. *Transform Lang Lit Technol Overv Learn* 2(2):9–15. <https://doi.org/10.55047/transtool.v2i2.1354>
- Taber KS (2018) The use of Cronbach's alpha when developing and reporting research instruments in science education. *Res Sci Educ* 48:1273–1296. <https://doi.org/10.1007/s11165-016-9602-2>
- UCLA Statistical Consulting Group (2020) What does Cronbach's alpha mean? UCLA: Statistical Consulting Group. <https://stats.idre.ucla.edu/spss/faq/what-does-cronbachs-alpha-mean/>
- Vera F (2023) Integrating Artificial Intelligence (AI) in the EFL classroom: Benefits and challenges. *Transformar* 4(2):66–77
- Wang X, Gao Y, Wang Q, Zhang P (2025) Fostering engagement in AI-assisted Chinese EFL Classrooms: The role of classroom climate, AI Literacy, and resilience. *Eur J Educ* 60(1):e12874. <https://doi.org/10.1111/ejed.12874>
- Wang Y, Xue L (2024) Using AI-driven chatbots to foster Chinese EFL students' academic engagement: An intervention study. *Comput Hum Behav* 159:108353. <https://doi.org/10.1016/j.chb.2024.108353>
- Wang Y, Xin Y, Chen L (2024) Navigating the emotional landscape: Insights into resilience, engagement, and burnout among Chinese High School English as a Foreign Language Learners. *Learn Motiv* 86:101978
- Wang Y, Wang Y, Pan Z, Ortega-Martin JL (2024) The predicting role of EFL students' achievement emotions and technological self-efficacy in their technology acceptance. *Asia-Pac Educ Res* 33(4):771–782. <https://doi.org/10.1007/s40299-023-00750-0>
- Wang D (2024) Teacher- Versus AI-Generated (Poe Application) Corrective Feedback and Language Learners' Writing Anxiety, Complexity, Fluency, and Accuracy. *Int Rev Res Open Distributed Learn*, 25(3). <https://doi.org/10.19173/irrodl.v25i3.7646>
- Warschauer M, Tseng W, Yim S, Webster T, Jacob S, Du Q, Tate T (2023) The affordances and contradictions of AI-generated text for writers of English as a second or foreign language. *J Second Lang Writ* 62(1):1–7. <https://doi.org/10.1016/j.jslw.2023.101071>
- Warschauer M, Xu Y (2024) Artificial intelligence for language learning: Entering a new era
- Wei L (2023) Artificial intelligence in language instruction: impact on English learning achievement, L2 motivation, and self-regulated learning. *Front Psychol* 14:1261955. <https://doi.org/10.3389/fpsyg.2023.1261955>
- Woo DJ, Susanto H, Yeung CH, Guo K, Fung AKY (2024) Exploring AI-Generated text in student writing: How does AI help? *Lang Learn Technol* 28(2):183–209
- Wordpress (2023) Channel View Publications and Multilingual Matters. Retrieved from: <https://channelviewpublications.wordpress.com/2023/07/28/five-tips-for-promoting-reflection-on-language-learning/#:~:text=Why%20is%20reflection%20important%3F,and%20sustain%20motivation%20for%20learning>
- Xia Q, Chiu TK, Chai CS (2022) The moderating effects of gender and need satisfaction on self-regulated learning through artificial intelligence (AI). *Educ Inf Technol* 28:8691–8713. <https://doi.org/10.1007/s10639-022-11547-x>
- Xu W, Meng J, Raja SKS, Priya MP, Kiruthiga Devi M (2023) Artificial intelligence in constructing personalized and accurate feedback systems for students. *Int J Model Simul, Sci Comput* 14(01):2341001
- Xu X, Dugdale DM, Wei X, Mi W (2023) Leveraging artificial intelligence to predict young learner online learning engagement. *Am J Distance Educ* 37(3):185–198. <https://doi.org/10.1080/08923647.2022.2044663>
- Xue Y (2022) The role of EFL teachers' self-efficacy and emotional resilience in appraisal of learners' success. *Front Psychol* 12:817388
- Yaghoobi M, Razmjoo SA (2016) The potentiality of computer-assisted instruction towards ameliorating Iranian EFL learners' reading level. *Computers Hum Behav* 59:108–114. <https://doi.org/10.1016/j.chb.2016.01.033>
- Yan D (2023) Impact of Chat GPT on learners in a L2 writing practicum: an exploratory investigation. *Educ Inf Technol* 1-25:1–25. <https://doi.org/10.1007/s10639-023-11742-4>
- Yang T (2024) Impact of Artificial Intelligence Software on English Learning Motivation and Achievement. In *SHS Web of Conferences* (Vol. 193, p. 02011). EDP Sciences. <https://doi.org/10.1051/shsconf/202419302011>
- Yuan L, Liu X (2024) The Effect of Artificial Intelligence Tools on EFL Learners' Engagement, Enjoyment, and Motivation. *Computers in Human Behavior*, 108474. <https://doi.org/10.1016/j.chb.2024.108474>
- Zhang C, Meng Y, Xinyu M (2024) Artificial intelligence in EFL speaking: Impact on enjoyment, anxiety, and willingness to communicate. *System* 121:103259. <https://doi.org/10.1016/j.system.2024.103259>
- Zhang R, Zou D (2020) Types, purposes, and effectiveness of state-of-the-art technologies for second and foreign language learning. *Comput Assist Lang Learn* 35:696–742. <https://doi.org/10.1080/09588221.2020.1744666>
- Zou B, Liviero S, Hao M, Wei C (2020) Artificial intelligence technology for EAP speaking skills: Student perceptions of opportunities and challenges. In M. R. Freiermuth, N Zarrinabadi (Eds.). *Technology and the psychology of second language learners and users*, 433–463. Palgrave Macmillan. https://doi.org/10.1007/978-3-030-34212-8_17

Acknowledgements

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Author contributions

MH designed the study, conceptualized the research, wrote the first draft, and supervised the project; BNA contributed to the methodology, data analysis, and interpretation of the results; JH assisted with data collection and provided critical revisions to the manuscript. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Ethics approval

This study involved human participants and their data. All procedures performed were in accordance with the ethical standards of the institutional and/or national research committee and the 1964 Declaration of Helsinki, including its later amendments. Ethical approval was granted by the Ethics Committee of the School of International Studies, Shaanxi Normal University, Xi'an, Shaanxi, China (Approval Number: SNU-2024-09-01), approved on September 1, 2024.

Informed consent

The survey was administered between 5 and 17 September 2024. At the outset of the survey, participants were provided with an introductory statement outlining the purpose of the study, the voluntary nature of participation, and the assurance of confidentiality. They were informed that there were no right or wrong answers and that their responses would be used solely for research purposes. Consent was obtained by participants' decision to proceed with the survey after reading this information.

Additional information

Correspondence and requests for materials should be addressed to Min He.

Reprints and permission information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025